

Systematic Review

Evolution of Risk Analysis Approaches in Construction Disasters: A Systematic Review of Construction Accidents from 2010 to 2025

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Abstract

Structural collapses are a major threat to urban safety and infrastructure resilience and as such there is growing research interest in understanding the causes and improving the prediction of risk to prevent human and material losses. Whether caused by fires, earthquakes or progressive failures due to overloads and displacements, these events have been the focus of investigation over the past 15 years. This systematic literature review looks at the use of formal risk analysis models in structural failures between 2010 and 2025 to map methodological trends, assess model effectiveness and identify future research pathways. From an initial database of 139 documented collapse incidents, only 42 were investigated using structured risk analysis frameworks. A systematic screening of 417 related publications yielded 101 peer-reviewed studies that met our inclusion criteria—specifically, the application of a formal analytical model. This discrepancy highlights a significant gap between the occurrence of structural failures and the use of rigorous, model-based investigation methods. The review shows a clear shift from single-method approaches (e.g., Fault Tree Analysis (FTA) or Finite Element Analysis (FEA)) to hybrid, integrated models that combine computational, qualitative and data-driven techniques. This reflects the growing recognition of structural failures as socio-technical phenomena that require multi-methodological analysis. A key contribution is the development of a strategic framework that classifies models by complexity, data requirements and cost based on patterns observed across the reviewed papers. This framework can be used as a practical decision support tool for researchers and practitioners to select the right model for the context and highlight the strengths and limitations of the existing approaches. The findings show that the future of structural safety is not about one single “best” model but about intelligent integration of complementary context-specific methods. This review will inform future practice by showing how different models can be combined to improve the depth, accuracy and applicability of structural failure investigations.

Keywords: collapse; accident analysis; risk management; structural analysis; risk analysis approach



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1. Introduction

Structural failures and construction accidents have caused significant human, economic, and environmental losses over the past few decades. Understanding the causes and mechanisms of these events has become a major focus in civil engineering and risk

management research. Various analytical approaches have emerged to investigate these disasters, ranging from traditional risk assessment frameworks to advanced computational simulations. Newman [1] proposed a risk management framework to identify and fix gaps in current practice, questioning underlying assumptions and addressing blind zones and vicious cycles to prevent future disasters. This foundational approach highlights the importance of critical thinking in risk analysis, especially when dealing with complex systems.

Other studies have focused on human factors, identifying lack of training and experience as major contributors to structural failures [2]. These human-centered studies show that risk is multidimensional and extends beyond technical design to organizational and behavioral aspects.

Several researchers have used forensic engineering methods to identify failure mechanisms. For example, the forensic geotechnical approach has been used to identify specific failure modes and recommend remedial measures to enhance the safety and stability of buildings in urban areas [3]. Post-earthquake collapses studies have used displacement-based frameworks to estimate structural damage under recorded ground motions, providing valuable insights into seismic vulnerability [4,5]. These studies emphasize the need to compare analytical results with theoretical methods to ensure the robustness and objectivity of findings [6].

Numerical and computational models have become increasingly prominent in accident analysis. To assess the progressive collapse susceptibility of typical garment factory buildings in Bangladesh, researchers used finite element analysis (FEA), which enables detailed and quantitative evaluation of structural behavior—essential for accurate risk assessment [7]. In another case, the U.S. Federal Highway Administration developed the HYRISK model, which has been applied in various disaster investigations, including the collapse of the I-5 Skagit River Bridge [8,9]. Computational simulation models have also been used to analyze collapse mechanisms due to overheight impacts, providing high specificity in understanding structural response [10]. Reliability-based models have further advanced the field. Cook et al. [11] developed a reliability-based risk analysis model to assess hazards to U.S. bridges under climate change, focusing on wind loads and scour depths while accounting for uncertainties in these estimates. Ghasemi [12] applied a system-level reliability model to investigate the Francis Scott Key Bridge collapse and found a critical misalignment between component and system reliability. This model shows how underestimating structural redundancy and misapplying load modifiers can lead to catastrophic failure. By combining structural criticality and design code principles, this approach offers a new perspective on risk assessment [12].

However, not all studies employ advanced modeling techniques. Some researchers have limited their analysis to listing causal factors without developing or applying formal risk models. For instance, a qualitative, literature-based, and observational review of building collapses in Nigeria relied on unreliable sources and did not involve technical or structural analysis, highlighting the gap between descriptive and analytical approaches [13]. In contrast, Caicedo et al. [14] used a joint structure–foundation–soil numerical model to analyze the same event, identifying inadequate column capacity as the primary cause, with secondary factors including differential settlements, high compressive stresses, and recent repair works. This contrast illustrates the evolution and necessity of moving from descriptive to analytical approaches in accident investigation.

In cases where predictive modeling is lacking, researchers have sought to fill the gap using innovative methods. The absence of predictive frameworks in studies on Nigerian building collapses led to the application of a comparative machine learning modeling approach, suitable for forecasting, classification, and identifying causal features when

sufficient data are available [15]. This shift toward data-driven models reflects a broader trend in risk analysis. Recent advances highlight the growing role of machine learning in predictive safety. A machine learning-based framework has been developed to enable real-time seismic damage prediction for earthquake early warning systems, demonstrating high accuracy in forecasting structural response [16], and also machine learning algorithms have also shown promise in predicting the damage of reinforced concrete frames under single and multiple seismic events, with applications deployed in user-friendly web platforms [17].

Hybrid and integrated models have also gained traction. Weiping et al. [18] combined fault tree analysis with Bayesian network methods to assess the post-earthquake functionality of subway stations. Similarly, other researchers have integrated 3D-FDM (Three-Dimensional Finite Difference Method) with FLAC 3D software to measure differential deformations in metro tunnels caused by earthquakes [19]. In the analysis of steel building construction projects, Leu [20] applied a Bayesian-network-based safety risk assessment model to evaluate object collapse hazards. These integrative approaches demonstrate the growing trend of combining models to enhance accuracy and comprehensiveness.

Natural and extreme events have also driven model development. The 2019 Brumadinho tailings dam collapse in Brazil—the country's worst human and environmental disaster—was analyzed using a multi-method remote sensing and geospatial framework to assess successive hazards and potential causes [21]. Meanwhile, hurricane and wind-induced collapses have been studied using hybrid models such as the CFD-FE (Computational Fluid Dynamics–Finite Element) model, which simulates high wind speeds on structures and evaluates their dynamic response [22].

Furthermore, construction work accidents—which extend beyond structural collapses—are equally critical. Widi et al. [23] analyzed hazards and risk levels in bridge construction operations, particularly in Indonesia, where casualties have increased. They applied an integrated approach combining Work Breakdown Structure (WBS), Risk Breakdown Structure (RBS), Analytic Hierarchy Process (AHP), and rating methods to achieve a comprehensive risk assessment [23]. This highlights the broader scope of risk analysis in construction, extending beyond structural integrity to operational safety.

This paper presents a systematic literature review on the evolution of risk analysis approaches used in the investigation of construction-related accidents and structural collapses between 2010 and 2025. It aims to enhance both academic and practical understanding of how risk management models have been applied to interpret real-world failures in the built environment. Unlike previous studies that focus on isolated case analyses or specific types of disasters, this review concentrates on a broad range of structural collapse events and the analytical frameworks used to investigate them.

The study examines peer-reviewed publications that address accident analysis in civil engineering contexts, with a focus on methodological rigor and the role of structural analysis in understanding failure mechanisms. It traces how risk analysis approaches have evolved over the past 15 years, highlighting shifts from descriptive assessments to advanced computational and integrated models. By mapping the development and application of these approaches, the review supports researchers and practitioners in identifying suitable methodologies for analyzing construction failures and improving safety outcomes. This paper does not assess the technical accuracy of individual models but instead provides a comprehensive overview of their use, context, and progression within the field of structural safety and risk management.

It is important to clarify that the dataset is organized around publications, not incidents. While 139 distinct collapse events were identified in the literature, only a subset of them ($n = 42$) were analyzed using formal risk models, resulting in 101 peer-reviewed articles

that met our inclusion criteria. Here, it is worth noting that this is a systematic literature review and not an empirical study; therefore, no primary data collection, model testing, or long-term monitoring implementation was conducted.

The purpose of this study is to map, analyze and critically evaluate the risk analysis models used in structural collapse investigations between 2010 and 2025. It has three objectives: (1) to identify and categorize the most used models; (2) to trace their methodological evolution; and (3) to develop a framework for model selection based on complexity, data needs and cost. This study was conducted to bridge the gap between methodological research and practical application, to provide a structured, evidence-based resource for researchers and practitioners.

Unlike previous reviews that focus on specific disaster types (e.g., earthquakes [4] or fires [24]) or isolated methodologies (e.g., FTA [25] or FEA [7]), this study provides a cross-domain synthesis of over 100 risk analysis models, so a comparative evaluation of their strengths, limitations and contextual applicability.

2. Review Methodology

2.1. Search Strategy and Screening Protocol

A systematic literature search was conducted following a methodology aligned with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The complete PRISMA Flow Diagram is available as Supplementary Figure S1, and the completed checklist is provided as Supplementary Table S1. The primary data sources were Scopus, Web of Science, and Engineering Village, selected for their comprehensive coverage of civil and structural engineering literature.

The search query combined keywords related to structural failures (e.g., “structural collapse”, “building collapse”, “bridge failure”) with terms for risk analysis models (e.g., “finite element analysis”, “fault tree analysis”, “STAMP”, “machine learning”). It is important to note that structural collapse incidents are often referred to by multiple names across publications. For instance, the ‘Morandi Bridge collapse’ is also known as the ‘Genoa bridge collapse’ or ‘Polcevera Viaduct collapse,’ while the ‘Plasco Building disaster’ appears as ‘Plasco Tower fire’ or ‘Tehran high-rise fire.’ Our search strategy accounted for such variations through a broad set of keywords and iterative screening. Boolean operators were used to refine results. No language restrictions were applied, but only studies published between 2010 and 2025 were included.

The screening process followed a three-stage approach:

1. Title and Abstract Screening: Initial filtering based on relevance to structural collapses and risk modeling.
2. Full-Text Assessment: Detailed evaluation against inclusion/exclusion criteria.
3. Quality Appraisal: Assessment of methodological rigor, focusing on the explicit use and description of a formal risk analysis model.

Inclusion Criteria:

- Empirical or case study-based research.
- Application of a structured risk analysis model to a specific structural collapse incident.
- Publication in a peer-reviewed journal.

Exclusion Criteria:

- General safety reviews without model application.
- Descriptive case reports lacking analytical frameworks.
- Conference papers (unless they presented novel, peer-reviewed methodologies).

This rigorous process ensured the final dataset comprised 101 high-quality, methodologically relevant studies suitable for synthesis and critical analysis.

2.2. Research Questions

Here are the three research questions that guide this systematic literature review, formulated to address the evolution, application, and integration of risk analysis approaches in construction-related accidents and structural collapses between 2010 and 2025:

RQ1. *What types of structural collapses and construction accidents have been investigated in the literature, and which risk analysis approaches were most commonly applied to each type of incident?*

This question aims to identify the scope of documented events, including buildings, bridges, dams, and industrial structures, and to map the dominant analytical models used in their investigation.

RQ2. *How have risk analysis approaches evolved between 2010 and 2025 in the context of structural failure investigations, and what are the strengths, limitations, and practical effectiveness of these methods?*

This question explores the methodological progression—from descriptive assessments to computational and hybrid models—and evaluates whether these approaches have achieved their intended purpose in improving safety and understanding failure mechanisms.

RQ3. *How has the integration of multiple risk analysis models influenced the accuracy and comprehensiveness of accident investigations, and under what conditions does model combination add value?*

This question examines the trend of combining frameworks—such as Bayesian networks with finite element analysis or fault tree analysis with geospatial tools—and assesses the added benefit of hybrid approaches in complex collapse scenarios.

2.3. Literature Search

In the field of accident science and risk analysis, conducting a systematic literature review (SLR) requires a structured and reproducible approach to identify relevant studies. This review followed a three-phase search strategy to ensure comprehensive coverage of publications related to structural collapses and construction accidents between 2010 and 2025.

- Phase 1: Identification of Key Accidents and Disasters

The first phase involved compiling a preliminary list of significant structural failures and construction-related disasters that occurred between 2010 and 2025. This step used open-source information from Google searches, accident databases, Wikipedia, and official incident reports. The focus was on events involving building or infrastructure collapse due to causes such as earthquakes, design flaws, material failure, fire, or human error. Only incidents with documented technical investigations or peer-reviewed analyses were retained. This phase resulted in a curated list of 42 major collapse events, including failures of buildings, bridges, dams, and industrial facilities.

- Phase 2: Database Search and Article Retrieval

The second phase focused on retrieving academic publications related to the identified accidents. Searches were conducted using three widely used digital libraries: Scopus, ScienceDirect, and Google Scholar. Scopus was selected as the primary database due to its extensive coverage of engineering records and its advanced search filters, which support reproducible and targeted queries. For each accident, a dedicated search was performed using the event's name (e.g., "Brumadinho dam collapse", "I-5 Skagit River Bridge collapse") and variations of the name to account for different spellings or reporting

styles. Keywords such as collapse, structural failure, accident analysis, risk management, and structural analysis were combined with the incident name to enhance retrieval accuracy.

To ensure completeness, forward and backward citation tracking was applied to key papers retrieved from Scopus. This process included checking reference lists (backward search) and citing articles (forward search) to capture relevant studies indexed in other databases, such as ScienceDirect and Google Scholar. Articles not initially found in Scopus but identified through these methods were added if they met the inclusion criteria. Scopus AI was also used to suggest additional relevant publications based on semantic matching, improving the efficiency of the search process.

- Phase 3: Screening and Relevance Assessment

In the third phase, the titles, abstracts, and, when necessary, full texts of all retrieved publications were reviewed. Each article was evaluated based on three key questions:

1. Does the study explicitly address the selected disaster and analyze the errors or root causes behind it?
2. Does it apply a specific model or framework for risk analysis, accident interpretation, or structural assessment?
3. Is the article suitable for a review focusing on risk analysis approaches in the construction industry, with a historical perspective from 2010 to 2025?

The screening process included both descriptive studies that identify causal factors without formal modeling and analytical studies that apply structured risk analysis approaches—such as deterministic models, probabilistic methods, computational simulations, or hybrid frameworks. The inclusion of descriptive studies allows this review to trace the evolution of accident analysis, from early qualitative assessments to more advanced, model-based investigations. However, the focus remains on identifying and evaluating the methodological rigor of studies that use formal risk analysis approaches, to understand how modeling practices have developed over time and contributed to improved structural safety.

It is worth noting that this review also includes studies that analyze general construction-related accidents or recurring collapse patterns of specific structure types within defined geographical regions, provided they apply formal risk analysis approaches. These studies contribute valuable insights into regional risk trends and the applicability of models across similar structural and environmental contexts.

This approach ensures that the final corpus reflects the full spectrum of analytical practices in the literature, while enabling a critical comparison between descriptive and model-driven methodologies.

The entire search process is illustrated in Figure 1, which outlines the sequential steps from accident identification to final study selection.

2.4. Literature Search Results

Following the identification of 139 structural failures and construction-related disasters that occurred between 2010 and 2025 through open-source information, a comprehensive literature search was conducted using the three academic databases specified in Section 2.2, Scopus, ScienceDirect, and Google Scholar. The search strategy, illustrated in Figure 2, guided the retrieval of peer-reviewed publications related to these incidents. Forward and backward citation tracking was also applied to ensure the inclusion of all relevant studies.

The search process yielded a total of 417 publications. After a systematic and thorough review, these were categorized as follows:

1. 202 publications were not related to any of the identified collapse events.
2. 59 publications discussed the incidents but did not apply a formal risk analysis model.

3. 33 publications were deemed unreliable due to insufficient methodological detail or lack of peer review.
4. 14 publications could not be accessed in full text.

This resulted in a preliminary set of 109 publications that addressed the topic of the review and were directly linked to risk analysis applications in structural collapse investigations. These publications focused on 42 distinct incidents, representing a subset of the initial 139, and were selected for in-depth analysis.

Only peer-reviewed journal articles, conference papers, books, and book chapters published by reputable sources were considered as high-quality references to ensure the validity and credibility of the review. The inclusion of these sources is summarized in Table 1.

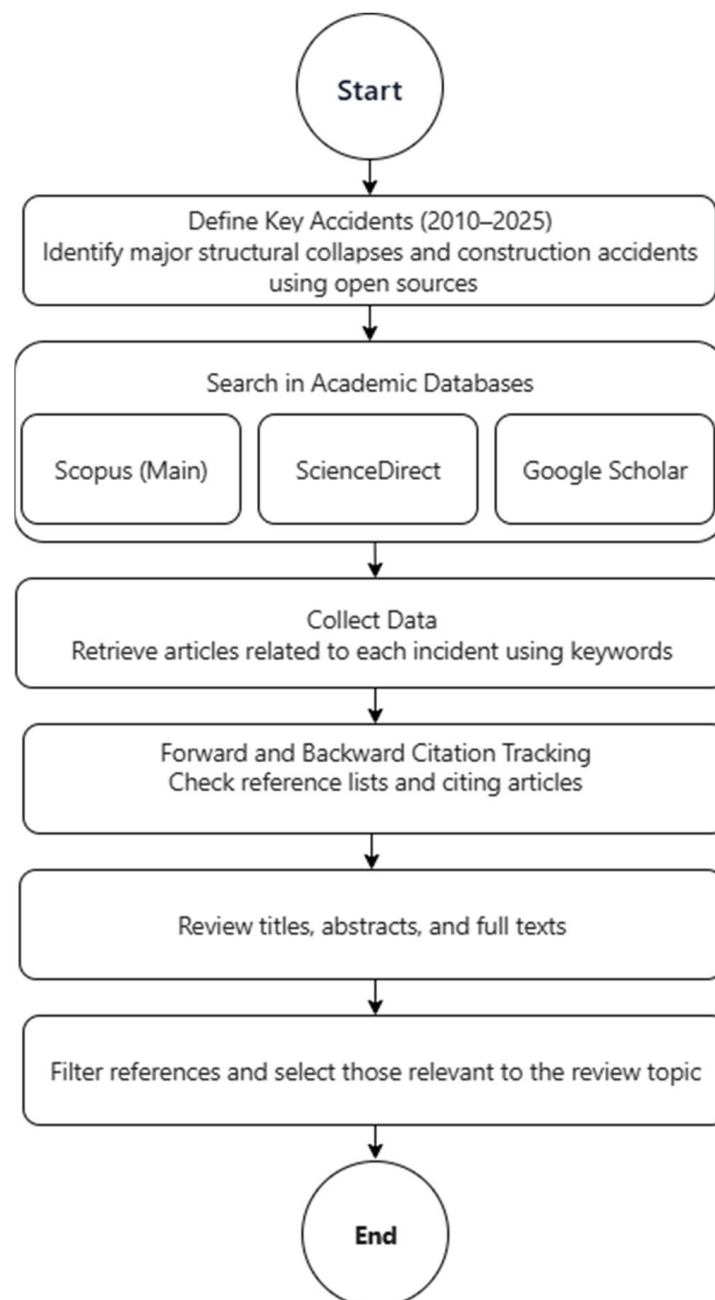


Figure 1. Search script flow diagram.

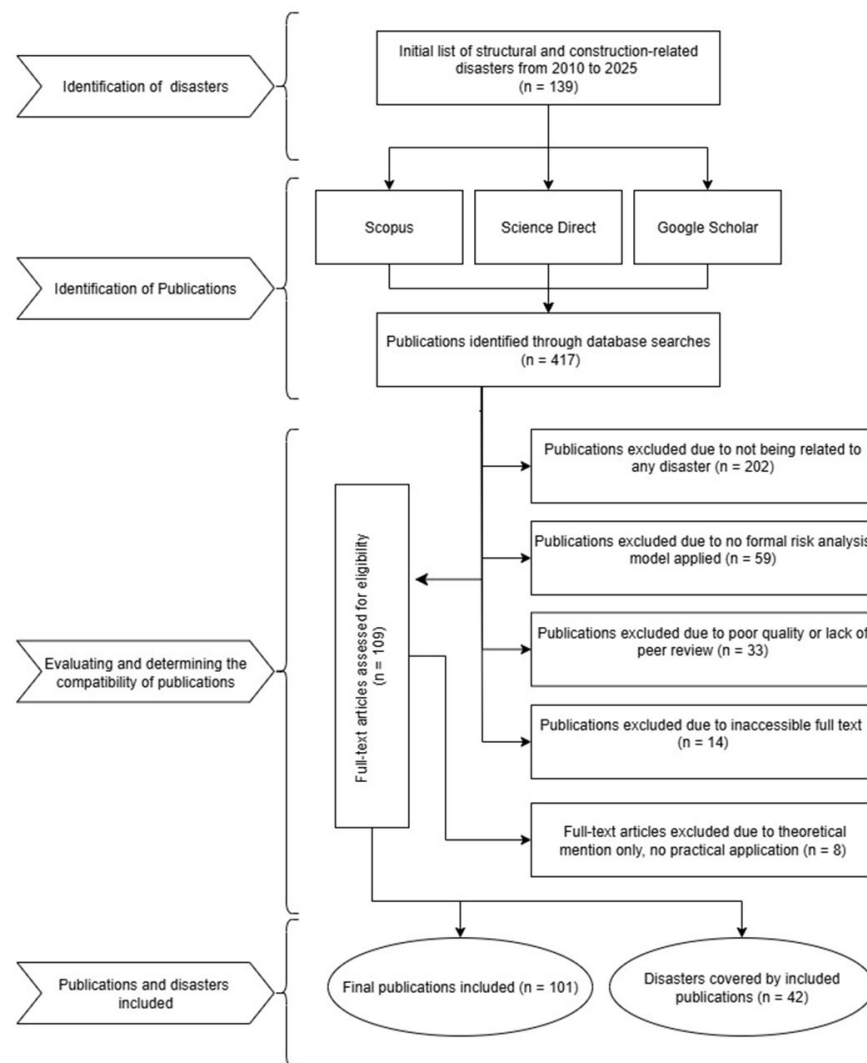


Figure 2. Literature search flow diagram.

Table 1. Publications related to the review.

Citation	Article Title	Year	Disaster	Risk Analysis Model Used
[1]	Beneath the horizon	2010	Deepwater Horizon	Blind Zones & Vicious Cycles
[5]	Seismic performance of reinforced concrete buildings in the 22 February Christchurch (Lyttleton) earthquake	2011	PGC building collapse	linear and nonlinear analysis of reinforced concrete buildings ASCE/SEI
[26]	The Indiana state fair collapse incident: Anatomy of a failure	2013	Indiana State Fair stage collapse	non-linear Finite Element Analysis (FEA) model
[4]	Performance-Based Issues from the 22 February 2011 Christchurch Earthquake	2013	Christchurch buildings	Displacement-based framework
[20]	Bayesian-network-based safety risk assessment for steel construction projects	2013	Object collapse in steel building construction projects	A Bayesian-network-based safety risk assessment model developed by transforming fault tree (FT) analysis into Bayesian networks (BN)

Table 1. Cont.

Citation	Article Title	Year	Disaster	Risk Analysis Model Used
[27]	Towards BBN based risk modelling of process plants	2014	Deepwater Horizon	The Bayesian Belief Network (BBN)
[28]	Collapse of the roof of a football stadium	2014	De Grolsch Veste stadium roof collapse	Finite Element Analysis (FEA) model
[29]	Risk analysis of collapse during construction for a subway transfer station with large span and small clearance	2014	Collapse during the construction of a subway transfer station	The Analytic Hierarchy Process (AHP)
[2]	“Everything was fine”: An analysis of the drill crew’s situation awareness on Deepwater Horizon	2015	Deepwater Horizon	Driller’s Situation Awareness (DSA) model
[6]	Buckling analysis of arched structures using finite element analysis	2015	Metrodome roof collapse	non-linear Finite Element Analysis (FEA) model
[11]	Bridge failure rate	2015	I-5 Skagit River Bridge collapse	A reliability-based risk analysis model
[8]	I-5 skagit river bridge collapse review	2016	I-5 Skagit River Bridge collapse	The HYRISK model
[30]	Lack of dynamic leadership skills and human failure contribution analysis to manage risk in Deepwaterdeep water horizon oil platform	2017	Deepwater Horizon	Fault Tree Analysis (FTA) Analytic Hierarchy Process (AHP)
[31]	Modeling progressive collapse of 2D reinforced concrete frames subject to column removal scenario	2017	Miami Dade College parking garage	A two-scale numerical model
[9]	Failing forward—construction failure case studies	2018	I-5 Skagit River Bridge collapse	the HYRISK model
[32]	Changes in land use and land cover as a result of the failure of a mining tailings dam in Mariana, MG, Brazil	2018	Mariana dam disaster	A multicriteria analysis approach (The Idrisi Selva [®] software)
[33]	Predicting buildings collapse due to seismic action in Lagos state	2018	Lagos building collapse	A Monte Carlo Simulation Model
[34]	Forecasting the hazards of seismic induced building collapse in Lagos-Nigeria through quality of reinforcing steel bars	2018	Lagos building collapse	A material-based risk assessment model
[35]	Preliminary modelling of Plasco Tower collapse	2018	Plasco Building	A 3D finite element model (OpenSees)
[36]	Technical and Administrative Assessment of Plasco Building Incident	2018	Plasco Building	A multi-perspective assessment approach
[37]	Considerations over the Italian road bridge infrastructure safety after the Polcevera viaduct collapse: past errors and future perspectives.	2018	Highway overpass collapse Italy	A comparative case study approach
[13]	Building collapse in south-south Nigeria	2018	Uyo Church collapse	A qualitative, literature-based, and observational review methodology

Table 1. Cont.

Citation	Article Title	Year	Disaster	Risk Analysis Model Used
[38]	Nonlinear dynamic analysis of the self-anchored suspension bridge subjected to sudden breakage of a hanger	2019	Bridge Collapse	Nonlinear dynamic analysis based on finite element theory
[14]	The collapse of Space building	2019	Collapse of the Space Building Colombia	A joint structure-foundation-soil numerical model
[39]	Critical infrastructures in Italy: State of the art, case studies, rational approaches to select the intervention priorities	2019	Highway overpass collapse Italy	RAM (Risk Assessment Model) with two key indices: Degree Relevance Index (IRD) Future Degradation Index (IDF)
[40]	Investigation of collapse of Florida International University (FIU) pedestrian bridge	2019	the Florida International University (FIU) pedestrian bridge collapse	A finite element (FE) numerical simulation model.
[41]	Collapse mechanism analysis of the FIU pedestrian bridge based on the improved structural vulnerability theory (ISVT)	2019	the Florida International University (FIU) pedestrian bridge collapse	The Improved Structural Vulnerability Theory (ISVT)
[42]	Pre-collapse space geodetic observations of critical infrastructure: The Morandi Bridge, Genoa, Italy	2019	Ponte Morandi collapse	A multi-geometry InSAR (Interferometric Synthetic Aperture Radar) analysis with MCMC (Markov Chain Monte Carlo) approach
[43]	Preliminary reflections on the failure of the Brumadinho tailings dam in January 2019	2019	Brumadinho dam disaster	A qualitative geotechnical assessment based on comparisons between historical cases and a review of historical records of dam safety.
[44]	Numerical Study on the Collapse of the Morandi Bridge	2020	Collapse of the Morandi Bridge	Finite Element Analysis (FEA) model
[45]	Effect of Load Cases and Hanger-Loss Scenarios on Dynamic Responses of the Self-Anchored Suspension Bridge to Abrupt Rupture of Hangers	2020	Bridge Collapse	The instantaneous stiffness degradation method
[46]	Exploring the collapse of buildings in urban settings	2020	Dar es Salaam building collapse	A probabilistic and modular approach
[7]	Assessment of Progressive Collapse Proneness of Existing Typical Garment Factory Buildings in Bangladesh	2020	Rana Plaza collapse	A finite element analysis (FEA) model
[47]	Risk priority number for bridge failures	2020	Vivekananda Flyover Bridge	The Failure Mode and Effect Analysis (FMEA) model
[48]	Evaluation of Plasco Building fire-induced progressive collapse	2020	Plasco Building	A three-phase approach: Field Investigation, Structural Evaluation, and Progressive Collapse Analysis

Table 1. Cont.

Citation	Article Title	Year	Disaster	Risk Analysis Model Used
[49]	Collapse of the 16-Story Plasco Building in Tehran due to Fire	2020	Plasco Building	A forensic engineering investigation
[3]	Investigation of Causes of Plasco Building Accident in Iran Using Timed MTO and ACCIMAP Methods: Investigation of Plasco 4 Building Accident in Iran	2020	Plasco Building	Timed MTO (TM) and Accimap methods
[50]	Monitoring and evaluation of bridges: lessons from the Polcevera Viaduct collapse in Italy	2020	Ponte Morandi collapse	Structural Health Monitoring (SHM)
[51]	Post-collapse analysis of Morandi's Polcevera viaduct in Genoa Italy	2020	Ponte Morandi collapse	A capacity-demand time-domain estimation model
[52]	Collapse analysis of the Polcevera viaduct by the applied element method	2020	Ponte Morandi collapse	The Applied Element Method (AEM)
[53]	A taxonomy of building collapse causes in Lagos State Nigeria	2020	Lagos school collapse	A hierarchical cluster analysis
[21]	The 2019 Brumadinho tailings dam collapse: Possible cause and impacts of the worst human and environmental disaster in Brazil	2020	Brumadinho dam disaster	A multi-method remote sensing and geospatial analysis framework
[54]	Seismic debris field for collapsed RC moment resisting frame buildings	2021	PGC building collapse	Applied Element Method (AEM) Deep Neural Network (DNN)
[10]	Overheight impact on bridges: A computational case study of the Skagit River bridge collapse	2021	I-5 Skagit River Bridge collapse	A computational simulation model
[55]	Risk dynamics modeling of reservoir dam break for safety control in the emergency response process	2021	Mariana dam disaster	The system dynamics model
[56]	Mining Disasters in Brazil: A Case Study of Dam Ruptures in Mariana and Brumadinho	2021	Mariana dam disaster	The Sendai Framework for Disaster Risk Reduction
[57]	Impact assessment of mechanical services to building instability	2021	Lagos building collapse	A field-based empirical analysis grounded in: Descriptive statistics, Chi-square tests, and Correlation analysis.
[58]	Failure assessment and virtual scenario reproduction of the progressive collapse of the FIU bridge	2021	the Florida International University (FIU) pedestrian bridge collapse	A finite element (FE) modeling approach (ABAQUS software)
[59]	Causes of the Collapse of the Polcevera Viaduct in Genoa, Italy	2021	Ponte Morandi collapse	A forensic engineering analysis
[60]	Watch out for the tailings pond, a sharp edge hanging over our heads: Lessons learned and perceptions from the brumadinho tailings dam failure disaster	2021	Brumadinho dam disaster	A multi-method remote sensing and spatiotemporal pattern analysis framework

Table 1. Cont.

Citation	Article Title	Year	Disaster	Risk Analysis Model Used
[61]	Construction safety during pandemics: Learning from the Xinjia express hotel collapse during COVID-19 in China	2021	Collapse of Xinjia Express Hotel	The Enhanced STAMP (Systems-Theoretic Accident Modeling and Processes) Model
[62]	Improving accident analysis in construction—Development of a contributing factor classification framework and evaluation of its validity and reliability	2021	Construction accidents in general	The Contributing Factors Classification System Framework (CFCSF) based on the Accimap method
[63]	Dynamic Response and Progressive Collapse of a Long-Span Suspension Bridge Induced by Suspender Loss	2022	Kutai Kartanegara Bridge Collapse	A fault-tree analysis (FTA) model
[64]	Constructing a Consumer-Focused Industry: Cracks, Cladding and Crisis in the Residential Construction Sector	2022	Thane building collapse	A Fault Tree Analysis (FTA) model
[65]	Modeling the collapse of the Plasco Building. Part I: Reconstruction of fire	2022	Plasco Building	A Computational Fluid Dynamics (CFD) fire modeling
[66]	Failure analysis of the 16-story Plasco building under re condition	2022	Plasco Building	A 3D nonlinear finite element analysis model (LS-DYNA software program)
[67]	Mechanisms Analysis for Fatal Accident Types Caused by Multiple Processes in the Workplace: Based on Accident Case in South Korea	2022	workplace accident mechanisms in South Korea	The Accident Mechanism of Multiple Processes (AMMP) model
[68]	Collapse analysis of the multi-span reinforced concrete arch bridge of Caprigliola, Italy	2022	Caprigliola bridge collapse	The Applied Element Method (AEM)
[19]	Failure analysis and deformation mechanism of segmented utility tunnels crossing ground fissure zones with different intersection angles	2022	Mexico City Metro collapse	A 3D-FDM (Three-Dimensional Finite Difference Method) with Flac 3D software
[69]	Safety risk estimation of construction project based on energy transfer model and system dynamics: A case study of collapse accident in China	2022	Fengcheng power station collapse	A hybrid approach combining the Energy Transfer Model (ETM) and System Dynamics (SD) theory
[70]	Data-driven determination of collapse accident patterns for the mitigation of safety risks at metro construction sites	2022	Metro construction collapse accidents from 1996 to 2021	Categorical data analysis
[71]	Progressive collapse assessment of Osmangazi Suspension Bridge due to sudden hanger breakage under different loading conditions	2023	Osmangazi Suspension Bridge	The progressive collapse analysis
[72]	Learning from the progressive collapse of buildings	2023	Miami Dade College parking garage	A Probabilistic Risk Assessment (PRA) model
[73]	Proactive Approach to Measure Safety Management on Building Projects in Saudi Arabia	2023	Mecca crane collapse	A multiple linear regression model

Table 1. Cont.

Citation	Article Title	Year	Disaster	Risk Analysis Model Used
[74]	Structural and Environmental Safety Studies of the Holy Mosque Area Using CFD	2023	Mecca crane collapse	A Computational Fluid Dynamics (CFD) model
[75]	Accident investigation and lessons not learned: accimap analysis of successive tailings dam collapses in Brazil	2023	Mariana dam disaster	The AcciMap analysis model
[76]	Towards Automation of Building Integrity Tracking: Review of Building Collapse in Nigeria	2023	Lagos building collapse	A systemic diagnostic framework combining: root cause analysis, a review-based analytical model, and a conceptual proposal for automating building
[77]	Fire modelling framework for investigating tall building fire: A case study of the Plasco Building	2023	Plasco Building	FITB (Fire Investigation for Tall Buildings)
[78]	Development of an agent-based model to improve emergency planning for floods and dam failures	2023	Brumadinho dam disaster	Life Safety Model (LSM)
[79]	A hybrid STAMP-fuzzy DEMATEL-ISM approach for analyzing the factors influencing building collapse accidents in China	2023	Collapse of Xinjia Express Hotel	A hybrid approach combining Systems-Theoretic Accident Modeling and Processes (STAMP) with triangular fuzzy Decision-Making Trial and Evaluation Laboratory (DEMATEL) and Interpretive Structural Modeling (ISM)
[80]	Anticipating the collapse of urban infrastructure: a methodology based on earth observation and MT-insar	2023	Mexico City Metro collapse	A methodology for monitoring and assessing the vulnerability of urban infrastructure using Earth Observation (EO) and Multi-Temporal Interferometric Synthetic Aperture Radar (MT-InSAR) time series analysis.
[81]	High-rise condominium collapse	2023	Champlain Towers South	A case study-based qualitative analysis using secondary data from news reports, public investigations, legal documents, and media coverage.
[82]	An analysis of the evolution of public sentiment and spatio-temporal dynamics regarding building collapse accidents based on Sina Weibo data	2023	Changsha building collapse	The Analytic Hierarchy Process (AHP)
[83]	Tretten Bridge: A numerical study into the collapse of a steel-glulam structure	2023	Tretten Bridge collapse	A numerical finite element analysis (FEA)-based methodology using linear static structural modeling.

Table 1. Cont.

Citation	Article Title	Year	Disaster	Risk Analysis Model Used
[84]	Failure analysis of the FC Twente Stadium roof collapse using the Swiss Cheese Accident Model	2023	Collapse of the roof at FC Twente's De Grolsch Veste stadium in Enschede	The Swiss Cheese Model (SCM)
[25]	Safety decision analysis of collapse accident based on "accident tree-analytic hierarchy process"	2023	General collapse accidents in construction	A combination of Fault Tree Analysis (FTA) and Analytic Hierarchy Process (AHP)
[85]	Reliability-Based Framework for Structural Robustness Evaluation of Bridges	2024	Bridge Collapse	A metamodel-based reliability analysis
[86]	Static and Dynamic Performance Analysis of Cable-Stayed Bridges with Cables Damaged Fire	2024	Fire Bridge Collapse	Finite element software and damage theory calculation methods
[87]	Investigation of the collapse of the Cincin lama bridge with consideration of fatigue damage	2024	The collapse of the Cincin Lama Bridge	Finite Element Analysis (FEA) model
[88]	Dynamic reliability analysis of Aerial Building Machine under extreme wind loads using improved QBDC-based active learning	2024	Mecca crane collapse	an improved Query-by-Dropout-Committee (QBDC)-based active learning (AL) approach combined with Deep Learning (DL)
[15]	Assessing the factors affecting building construction collapse casualty using machine learning techniques: a case of Lagos, Nigeria	2024	Lagos building collapse	A comparative machine learning modeling approach
[89]	The sustainable development of bridges in China: Collapse cause analysis, existing management dilemmas and potential solutions	2024	Chirajara bridge collapse	A classification approach to analyze bridge collapse causes globally
[90]	An LLM-Based Inventory Construction Framework of Urban Ground Collapse Events with Spatiotemporal Locations	2024	Zhejiang bridge collapse	large language models (LLMs) to extract and analyze data from news reports
[91]	Notre-Dame de Paris as a validation case to improve fire safety modelling in historic buildings	2024	Notre-Dame de Paris fire	The Fire Dynamics Simulator (FDS)
[18]	Post-earthquake functionality assessment of subway stations considering the interdependency among sub-systems	2024	Mexico City Metro collapse	A combination of fault tree analysis and Bayesian network methods
[92]	Geohazard assessment of Mexico City's Metro system derived from SAR interferometry observations	2024	Mexico City Metro collapse	A combination of satellite radar interferometry observations, leveling surveys, subsurface profiles, linear gradient and differential displacement analyses, and structural-engineering parameters.

Table 1. Cont.

Citation	Article Title	Year	Disaster	Risk Analysis Model Used
[93]	Dilemmas and Solutions for Sustainability-Based Engineering Ethics: Lessons Learned from the Collapse of a Self-Built House in Changsha, Hunan, China	2024	Changsha building collapse	An analysis and decision-making model based on the Civil Code of the People's Republic of China
[94]	Threats to the preservation of the vicerojal church facades of Juli, Peru	2024	Santa Cruz church collapse	A qualitative, multi-hazard risk assessment model based on a geographic information system (GIS) framework.
[95]	Evaluating the damage of collapsed bridges using remote sensing technologies: Case study: Baltimore's Francis Scott Key Bridge	2024	Francis Scott Key Bridge collapse	A post-event damage assessment methodology based on remote sensing and geospatial analysis.
[96]	Modification of HFACS model for path identification of causal factors of collapse accidents in the construction industry	2024	Fengcheng power station collapse	A modified version of the Human Factors Analysis and Classification System (HFACS)
[97]	The Synergy and Accumulation Model for Analysis (SAMA): A Novel Approach to Transforming Risk Analysis in Construction with a Focus on the Deepwater Horizon Disaster Case	2025	Deepwater Horizon	The Synergy and Accumulation Model for Analysis (SAMA)
[24]	Analysis of the Plasco tower in fire using an integrated simulation approach	2025	Plasco Building	Fire Dynamics Simulator (FDS) & OpenSEES
[98]	Collapse of the Chirajara Cable-Stayed Bridge in Colombia	2025	Chirajara bridge collapse	A combination of global analyzes and refined nonlinear analyses, involved: in situ inspection, analytical studies, review of construction documentation, and testing of materials.
[99]	Bridge collapses in Italy across the 21st century: survey and statistical analysis	2025	Ponte Morandi collapse	A statistical database approach combined with case study analysis
[100]	Sense listening and the Reorganization of Collective Action During Crisis Management: The Notre-Dame de Paris fire	2025	Notre-Dame de Paris fire	A sensemaking framework centered around the concept of "senselistingening"
[22]	Analysis of Hurricane Maria's Impact on the Arecibo Telescope	2025	Arecibo Telescope	A combined CFD-FE (Computational Fluid Dynamics-Finite Element) analysis model
[12]	Analyzing reliability missteps: The collapse of the Francis Scott Key Bridge—Target reliability, redundancy, and extreme load combinations	2025	Francis Scott Key Bridge collapse	A system level and element level reliability model.
[101]	Robustness-based assessment and monitoring of steel truss railway bridges to prevent progressive collapse	2025	Francis Scott Key Bridge collapse	A consequence-based robustness assessment model

Table 1. *Cont.*

Citation	Article Title	Year	Disaster	Risk Analysis Model Used
[102]	An Overview of the Main Types of Damage and the Retrofitting of Reinforced Concrete Bridges.	2025	Dresden Carola Bridge collapse	A qualitative, literature-based review methodology
[103]	China-Pakistan Belt and Road Joint Laboratory on Smart Disaster Prevention of Major Infrastructures	2025	2025 Bangkok skyscraper collapse	A qualitative, multidisciplinary case study approach
[23]	Safety Risk Assessment Model for Bridge Construction	2025	Bridge construction accidents in Indonesia	The Work Breakdown Structure (WBS), Risk Breakdown Structure (RBS), Analytic Hierarchy Process (AHP), and rating methods

Non-peer-reviewed materials were excluded. Furthermore, during the full-text evaluation, it was found that 8 of the 109 publications mentioned risk models only in a descriptive or theoretical manner, without demonstrating practical application to the incident under study. These were primarily forensic or statistical reports that listed causal factors but did not conduct structured model-based analysis. As such, they were excluded to maintain methodological rigor.

The final dataset consisted of 101 publications that explicitly applied a formal risk analysis approach—such as deterministic, probabilistic, computational, or hybrid models—to analyze the causes, mechanisms, or consequences of structural failures.

It is worth noting that the review included studies that analyzed construction-phase accidents using structured risk models, as well as those that examined collapse events of specific structure types using formal analytical frameworks. All included publications met the strict selection criteria outlined in Section 2.3, ensuring their relevance and methodological soundness.

2.5. Inclusion/Exclusion Criteria

To ensure the quality and relevance of the literature included in this systematic review, clear inclusion and exclusion criteria were applied during the screening process. The primary objective was to identify studies that contributed to understanding how risk analysis approaches have been used in the investigation of construction-related accidents and structural collapses between 2010 and 2025.

Inclusion Criteria

The following criteria were used to select publications for inclusion:

1. **Publication Type:** Peer-reviewed journal articles, conference papers, book chapters, books, and reviews were included. Gray literature such as technical reports and official investigation documents was considered only if they provided detailed methodological descriptions of risk models or accident analysis frameworks.
2. **Time Frame:** Only publications released between 2010 and 2025 were included to ensure coverage of recent developments in risk modeling and accident investigation.
3. **Subject Matter:** Studies must address real-world structural failures, including building collapses, bridge failures, dam breaches, or industrial facility disasters. The focus was on incidents related to construction, operation, or maintenance phases.

Analytical Approach: Publications were included if they explicitly applied a formal risk analysis approach—such as deterministic models, probabilistic methods, computational simulations, or hybrid frameworks—to interpret the causes, mechanisms, or con-

sequences of the collapse. Descriptive studies that listed failure factors without using a structured model were also included to trace the evolution of analytical practices, but their contribution was evaluated critically in terms of methodological rigor.

Exclusion Criteria

Publications were excluded if they met any of the following conditions:

1. They focused solely on theoretical or hypothetical scenarios without reference to actual collapse events.
2. They lacked sufficient detail on the risk analysis methodology used, making it impossible to assess its application or validity.
3. They were not written in English or did not provide accessible content in English translation.
4. They dealt with non-structural failures (e.g., electrical system faults, fire spread without structural collapse) unless they directly addressed structural integrity or collapse mechanisms.

After applying these criteria, a total of 109 publications were selected for in-depth analysis. These included 92 journal articles, 8 conference papers, 5 reviews, 2 research reports, 1 book, and 1 book chapter (see Figure 3). The majority of the selected publications were peer-reviewed journal articles, reflecting the dominance of academic research in this field. Conference proceedings and reviews contributed additional insights into emerging trends and synthesis of existing knowledge.

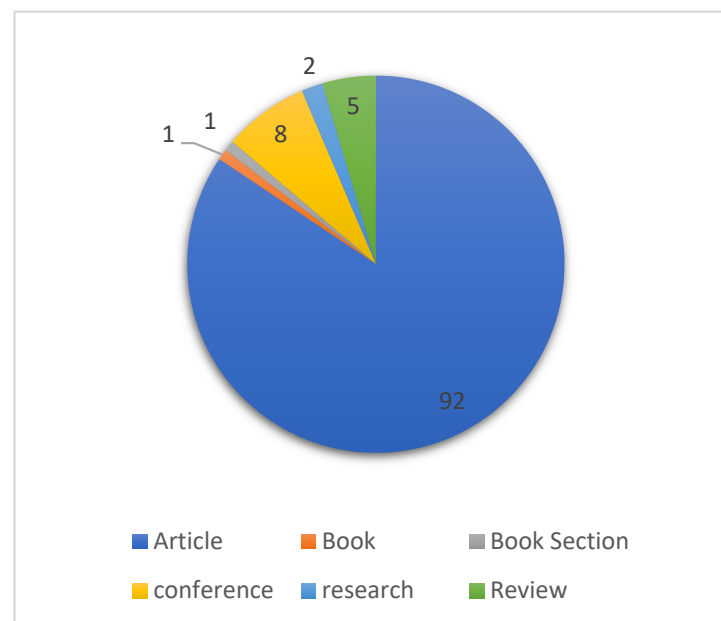


Figure 3. Distribution of publications by type.

This distribution highlights the strong presence of original research in the literature, while also indicating the growing role of comprehensive reviews and case-based analyses in advancing understanding of structural failure investigations.

3. Scientometric Analysis

3.1. Publication Source

Journal publications represent a primary channel for disseminating scientific knowledge in the field of structural safety and risk analysis. A scientometric analysis of the 101 included studies reveals a diverse distribution of publication sources in terms of journal ranking and publisher contribution.

- Journal Ranking (SJR)

The majority of the included articles were published in high-quality journals indexed in Scopus. As shown in Figure 4, the journals were categorized according to their Scimago Journal Rank (SJR) quartiles. Of the 101 publications, 57 (56.4%) were published in Q1-ranked journals, indicating a strong presence in top-tier scientific outlets. An additional 23 (22.7%) were published in Q2 journals, 8 (7.9%) in Q3, and 6 (5.9%) in Q4. The remaining 17 publications consisted of conference papers and research reports, which are also recognized as valuable contributions, particularly in engineering practice.

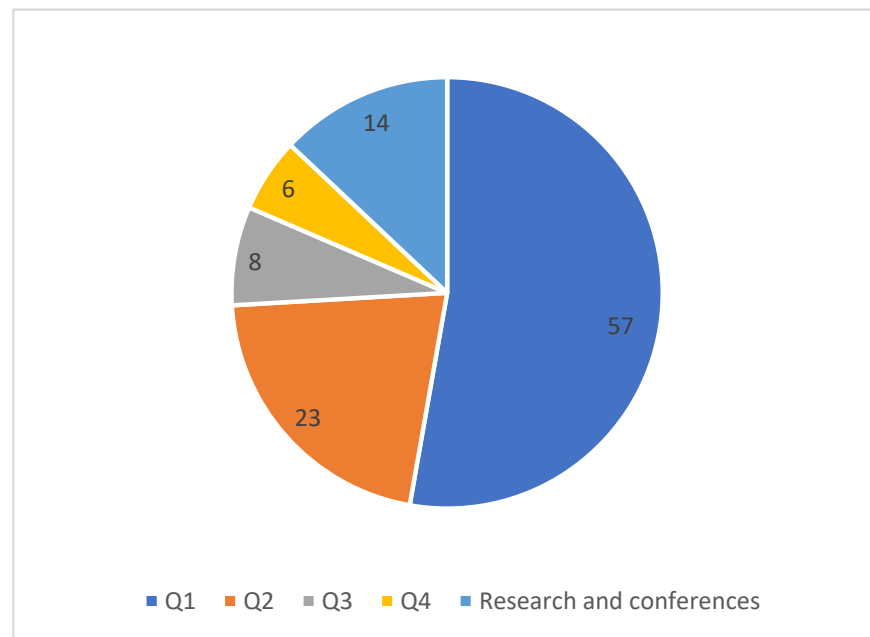


Figure 4. Journal classification of related publications.

This distribution highlights the academic rigor of the selected literature and reflects the growing interest in structural collapse investigations within high-impact journals.

- Publisher Contribution

Figure 5 presents the distribution of publications by publisher, illustrating the leading role of major academic publishers in disseminating research on construction-related accidents and risk modeling. The analysis shows that Elsevier B.V. is the leading publisher, contributing 18 articles (17.6%), followed by MDPI with 15 articles (14.7%), and the American Society of Civil Engineers (ASCE) with 13 articles (12.7%). Other notable contributors include Elsevier Ltd. (9), Springer Science + Business Media (4), and ICE Publishing (3). A total of 22 articles (22.5%) were published by other publishers, indicating a broad yet concentrated publishing landscape. The dominance of publishers such as Elsevier, MDPI, and ASCE underscores their pivotal role in advancing research on structural failure analysis, risk management, and accident investigation.

These findings suggest that research in this domain is not only methodologically robust but also widely disseminated through reputable and high-visibility publishing platforms.

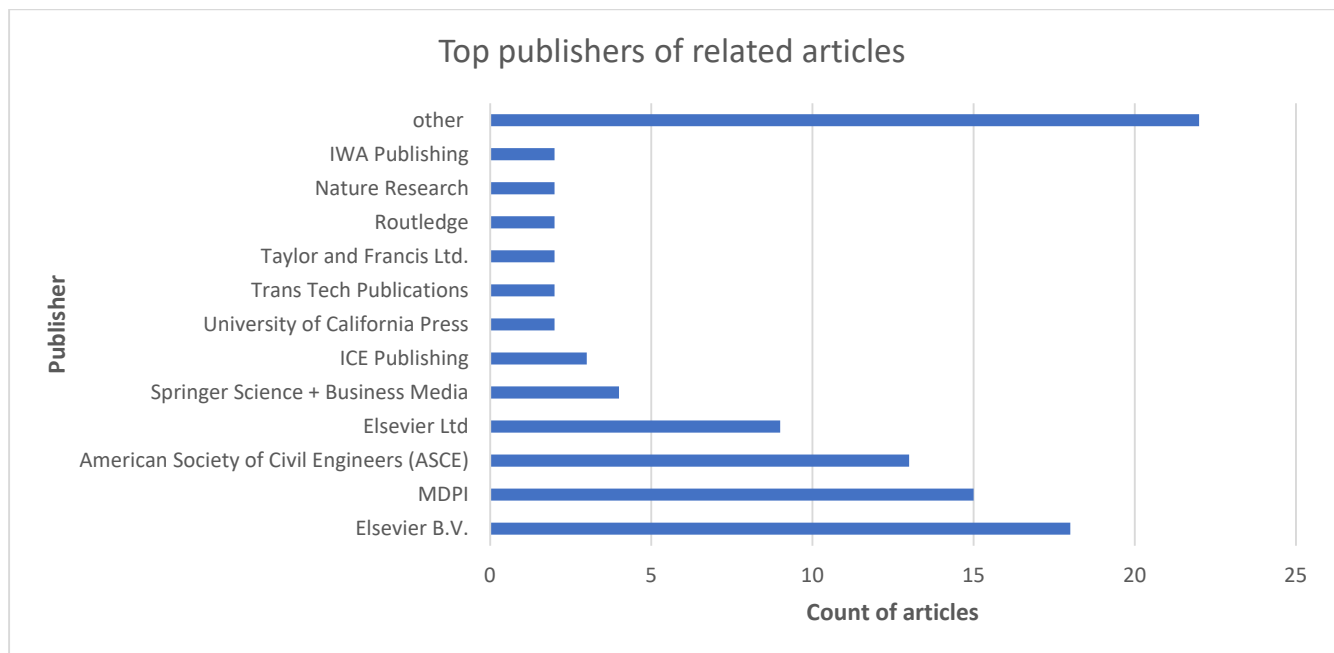


Figure 5. Included publication sources.

3.2. Most Cited Disasters

This section presents a scient metric analysis of the most frequently studied disasters in the literature, focusing on their type, geographic distribution, temporal trends, and publication impact. Of the 42 structural disasters included in this review, certain events have received significantly more scholarly attention due to their high human toll, engineering implications, or media visibility.

As shown in Table 2, the Plasco Building collapse in Tehran (2017) and the Morandi Bridge collapse in Italy (2018) are the most cited disasters, with 9 and 7 publications, respectively. These events not only caused significant loss of life (21 and 43 fatalities, respectively), but also raised critical questions about structural integrity, maintenance practices, and regulatory oversight.

Table 2. Disasters covered by the review.

Disaster	Number of Publications	Country	Year of Disaster	Casualties	Type
Lalita Park building collapse	1	India	2010	71 dead	Building
Deepwater Horizon	6	USA	2010	11 dead and catastrophic environmental damage	Offshore drilling rig
Metrodome roof collapse	1	USA	2010		Stadium
Kutai Kartanegara Bridge Collapse	1	Indonesia	2011	20 dead, 39 injured	Bridge
De Grolsch Veste stadium roof collapse	2	The Netherlands	2011	2 dead, 14 injured	stadium
PGC building collapse	3	New Zealand	2011	18 dead	Building
Indiana State Fair stage collapse	1	USA	2011	7 dead, 58 injured	Building
Miami Dade College parking garage	2	USA	2012	4 dead, 7 injured	Parking garage

Table 2. Cont.

Disaster	Number of Publications	Country	Year of Disaster	Casualties	Type
Rana Plaza collapse	1	Bangladesh	2013	1134 dead, +2500 injured	Building
Collapse of the Space Building Colombia	3	Colombia	2013	12 dead	Building
Thane building collapse	1	India	2013	74 dead, 60–62 injured	Building
Dar es Salaam building collapse	1	Tanzania	2013	36 dead, 18 injured	Building
I-5 Skagit River Bridge collapse	4	USA	2013	3 injured	Bridge
Mariana dam disaster	3	Brazil	2015	19 dead, 16 injured	Dam
Mecca crane collapse	3	Saudi Arabia	2015	118 dead, 394 injured	Crane
Lagos building collapse	5	Nigeria	2016	34 dead	Building
Uyo Church collapse	1	Nigeria	2016	60 dead	Church
Fengcheng power station collapse	3	China	2016	74 dead, 2 injured	Power station
Vivekananda Flyover Bridge	1	India	2016	27 dead, 80 injured	Bridge
Plasco Building	9	Iran	2017	21 dead, 70 injured, 1 missing	Building
Highway overpass collapse Italy	2	Italy	2017	2 dead, 3 injured	Highway overpass
Zhejiang bridge collapse	1	China	2018	8 dead, 3 injured	Bridge
Chirajara bridge collapse	2	Colombia	2018	9 dead, 5 injured	Bridge
The collapse of the Cincin Lama bridge	1	Indonesia	2018	27 dead, 10 injured	Bridge
Collapse of the Morandi Bridge	7	Italy	2018	43 dead, 16 injured	Bridge
the Florida International University (FIU) pedestrian bridge collapse	3	USA	2018	6 dead, 9 injured	Bridge
Lagos school collapse	1	Nigeria	2019	20 dead, over 60 injured	Building
Brumadinho dam disaster	5	Brazil	2019	270 dead	Dam
Notre-Dame de Paris fire	2	France	2019	3 injured	Church
Arecibo Telescope	1	Puerto Rico	2020	0 dead or injured	Telescope
Collapse of Xinjia Express Hotel	2	China	2020	29 dead, 42 injured	Building
Caprigliola bridge collapse	1	Italy	2020	2 injured	Bridge
Mexico City Metro collapse	4	Mexico	2021	26 dead, 79 injured	Metro tunnel
Champlain Towers South	2	USA	2021	98 dead, 11 injured	Building
Changsha building collapse	2	China	2022	54 dead, 10 injured	Building
Tretten Bridge collapse	1	Norway	2022	None	Bridge
Fern Hollow Bridge collapse	1	USA	2022	10 injured	Bridge

Table 2. *Cont.*

Disaster	Number of Publications	Country	Year of Disaster	Casualties	Type
Santa Cruz church collapse	1	Mexico	2023	11 dead, 60 injured	Church
Osmangazi Suspension Bridge	1	Turkey	2023		Bridge
Dresden Carola Bridge collapse	1	Germany	2024	No fatalities or injuries	Bridge
Francis Scott Key Bridge collapse	3	USA	2024	6 dead, 2 injured	Bridge
2025 Bangkok skyscraper collapse	1	Thailand	2025	53 dead, 9 injured, 41 missing	Skyscraper

The Deepwater Horizon oil rig disaster (2010), although not a traditional structural collapse, is also among the most analyzed, with 6 publications, due to its catastrophic environmental and operational consequences. Similarly, the Brumadinho (2019) and Mariana (2015) dam disasters in Brazil have drawn substantial academic interest, with 5 publications each, highlighting the growing focus on geotechnical safety and risk modeling in tailings dam engineering.

Other notable disasters with multiple studies include:

- Lagos building collapse (Nigeria, 2016)—5 publications
- Champlain Towers South (USA, 2021)—2 publications
- Francis Scott Key Bridge (USA, 2024)—3 publications
- Chirajara Bridge (Colombia, 2018)—2 publications

These events reflect a global pattern of structural failures, with a concentration in urban and high-density environments, where the consequences of collapse are most severe.

- Geographic Distribution

The disasters covered in the literature occurred across 21 countries, with the highest number reported in the United States (9 disasters), followed by China (4), India (3), Italy (3), and Nigeria (3). This distribution indicates that structural failures are a global phenomenon, but their documentation and academic analysis are more prevalent in countries with strong research infrastructure and accessible technical reporting systems. The concentration of studies in these regions may reflect both the frequency of incidents and the availability of detailed forensic investigations, rather than a higher inherent risk of collapse.

- Temporal Trends

The disasters span from 2010 to 2025, with a noticeable increase in both the number of incidents and the volume of academic publications after 2015. This rise correlates with growing public awareness, improved monitoring systems, and increased use of digital tools in forensic engineering.

- Disaster Type Distribution

As illustrated in Figure 6, bridge collapses ($n = 14$) are the most frequently studied type of disaster, followed by building collapses ($n = 13$). This reflects the complexity of bridge design, exposure to dynamic loads, and the high visibility of such failures. Other types, including dams, churches, stadiums, and offshore structures, are less common but still significant in specific contexts.

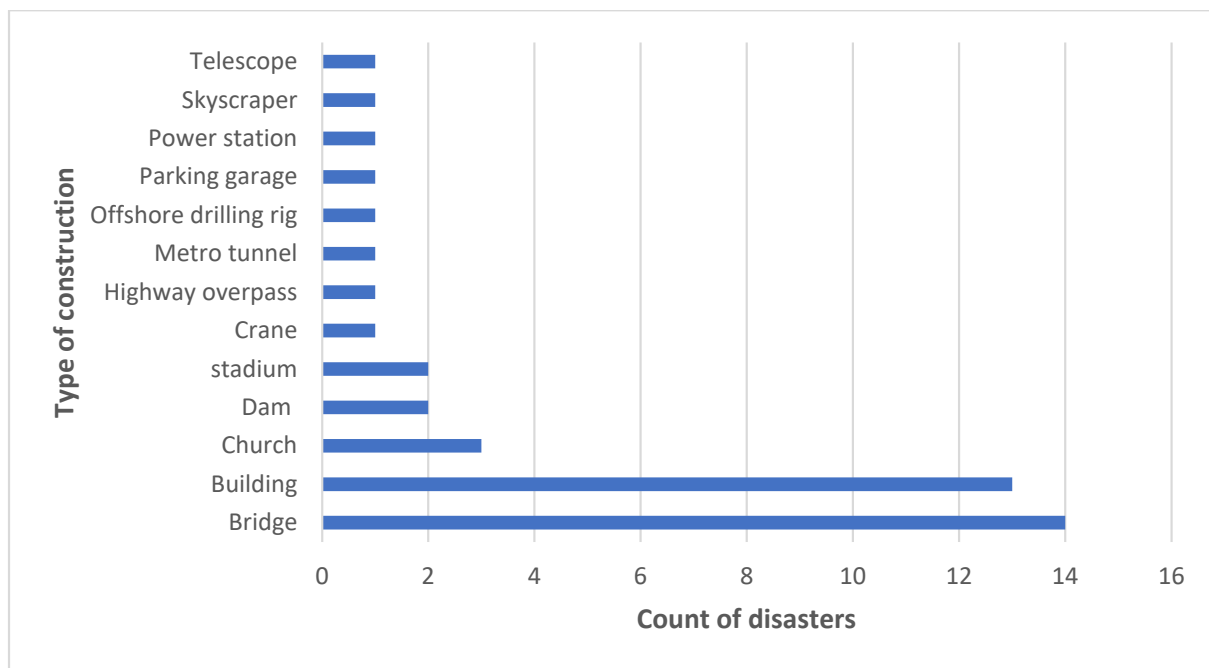


Figure 6. Types of disasters covered in the literature related.

The dominance of bridge and building collapses in the literature underscores the need for more robust risk analysis frameworks tailored to these structures, particularly in regions with aging infrastructure or rapid urban development.

- **Distribution of Hazard Types and Analysis of Compound Threats**

An analysis of the 101 studies reveals a wide range of structural collapse incidents, with structural failure and progressive collapse being the most common ($n = 14$), including Rana Plaza, FIU pedestrian bridge, Morandi Bridge, and Champlain Towers South collapses. This is followed by construction and operational accidents ($n = 12$), which include building and bridge collapses during construction or maintenance, such as Lalita Park, Thane, and Kutai Kartanegara Bridge. Geotechnical and water-related failures, including dam breaches and ground instability, were studied in 6 studies, including Mariana and Brumadinho tailings dam disasters and Mexico City Metro collapse.

Less common are fire-induced collapses ($n = 3$), such as Plasco Building and Notre-Dame de Paris fire, and extreme weather events ($n = 2$), including Arecibo Telescope collapse due to Hurricane Maria. Human and organizational factors (HOFs) were the main focus in 4 incident analyses, such as Deepwater Horizon, Xinjia Express Hotel, and Space Building collapse in Colombia.

Important to note is that more than 35 recent studies adopt a multi-threat approach, recognizing that structural collapses are rarely caused by a single factor. Instead, they result from the interaction of technical, human, organizational and environmental factors. For example, the Plasco Building collapse [24] was studied through fire dynamics (FDS), structural response (FEA) and systemic organizational failures (AcciMap). Similarly, the Deepwater Horizon disaster [1] was analyzed through blind zones in risk management, leadership failure and technological oversight.

This shift towards analyzing compound threats is a fundamental change in the field: modern risk analysis is moving beyond single-cause models to the complexity of real-world disasters. The prevalence of hybrid and integrated models—such as STAMP-fuzzy DEMATEL-ISM [79], FTA-BN [18], and FDS-OpenSEES [24]—supports this trend. These frameworks allow researchers and practitioners to trace how small, overlooked risks

accumulate and interact over time to produce catastrophic outcomes—a principle also embodied in emerging models like SAMA [97].

3.3. Most Cited Risk Analysis Models

This section identifies the most commonly used risk analysis models in the investigation of structural collapses and construction-related disasters between 2010 and 2025. Based on the 101 included publications, some models have been applied more frequently due to their flexibility, methodological clarity, or relevance to specific failure types.

The analysis shows that finite element analysis (FEA) is the most widely used modeling approach, appearing in 15 publications between 2013 and 2025. Originating in the early 1940s, FEA has become a cornerstone of structural simulation, enabling detailed assessment of stress distribution, deformation, and progressive collapse mechanisms. It is used for nonlinear dynamic analysis (e.g., OpenSees, ABAQUS), 3D modeling (LS-DYNA, FLAC 3D), and is often combined with software tools and damage theory. The continued use of FEA highlights its enduring relevance in forensic structural analysis.

Fault Tree Analysis (FTA), developed in 1962, is the second most cited framework, with 6 publications applying it directly or in hybrid forms. FTA has been used to map causal chains in bridge collapses, dam failures, and building accidents. Several studies have enhanced FTA by combining it with other methods, such as the Analytic Hierarchy Process (AHP) and Bayesian networks (BN), to improve uncertainty handling and decision-making under complex conditions.

Qualitative analysis approaches are also common, appearing in 19 publications between 2015 and 2025. These include forensic engineering investigations, case study-based reviews, and multi-perspective assessment frameworks. Although not computational, these methods are frequently used when data is limited or when the focus is on human and organizational factors. The increasing use of large language models (LLMs) and systemic diagnostic frameworks since 2023 indicates a growing trend toward integrating qualitative insights with data extraction technologies.

Other notable models include:

- Remote sensing and geospatial analysis (3 publications), particularly in dam and infrastructure monitoring using MT-InSAR and satellite radar.
- Bayesian Belief Networks (BBN) (3 publications), valued for probabilistic reasoning and risk propagation modeling.
- STAMP (Systems-Theoretic Accident Modeling and Processes) (2 publications), used to analyze systemic failures in complex socio-technical systems.
- The AcciMap model (3 publications), applied to classify contributing factors across organizational levels.
- Progressive collapse analysis (2 publications), often used in post-collapse structural evaluation.

Additionally, displacement-based frameworks and material-based risk assessment models have been applied in seismic and material failure contexts, while the Fire Dynamics Simulator (FDS) has been used in fire-induced collapse investigations.

The temporal distribution of these models shows a clear trend: while traditional methods like FTA and FEA remain dominant, recent years (2020–2025) have seen an increase in hybrid and data-driven approaches, such as metamodel-based reliability analysis, multi-hazard GIS frameworks, and AI-assisted data extraction. This evolution reflects a growing emphasis on combining physical modeling with data-rich environments to enhance the depth and scope of accident analysis.

In addition to the frequently applied models, several studies used unique or context-specific risk analysis frameworks only once in the reviewed literature. These include

the HYRISK model, reliability-based approaches, probabilistic risk assessment (PRA), computational fluid dynamics (CFD) fire modeling, and emerging methods such as machine learning-based active learning and the Sendai Framework for Disaster Risk Reduction. While these models lack widespread adoption, they reflect the diversity of analytical strategies and the adaptation of risk assessment tools to specific disaster types, materials, or regulatory environments.

3.4. Keywords Co-Occurrence

To explore the thematic structure of the literature on structural collapse investigations, a keyword co-occurrence analysis was conducted using VOSviewer 1.6.18 on the 101 included publications. The search strategy combined disaster-specific terms with core analytical keywords such as collapse, progressive collapse, risk analysis, cause analysis, causes, effects, and damage. This ensured that all studies addressing both physical failure mechanisms and the methodological frameworks used in accident interpretation were captured.

The analysis yielded 832 unique keywords, from which a network of 27 frequently occurring and thematically relevant terms was constructed. As shown in Figure 7, node size represents keyword frequency, and link thickness indicates the strength of co-occurrence between terms.

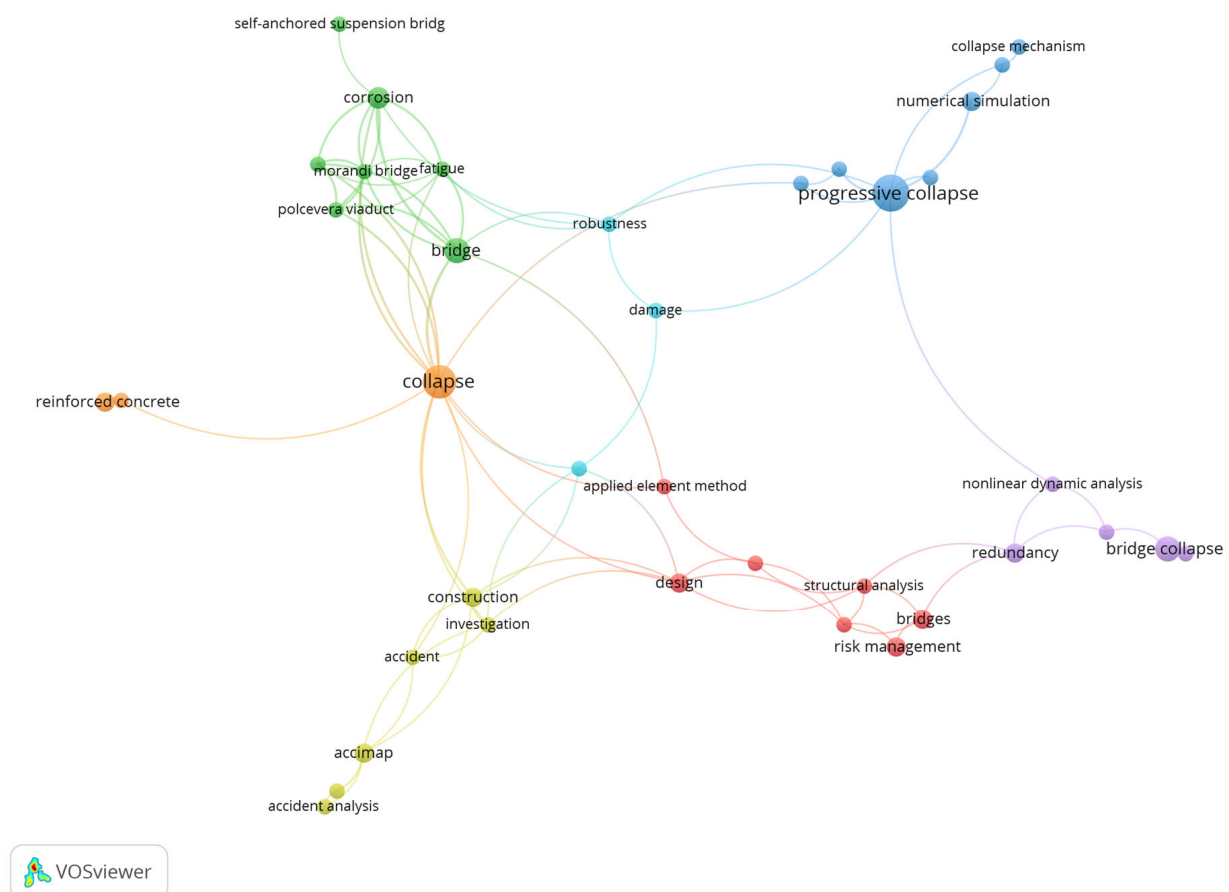


Figure 7. Keyword co-occurrence network.

The network reveals three main clusters:

- Red cluster: Focused on structural analysis, finite element analysis (FEA), numerical simulation, and nonlinear analysis—representing the computational and modeling dimension of collapse investigations.

- Green cluster: Centered on risk management, risk assessment, safety, and failure modes—highlighting the risk-oriented and preventive aspects of the research.
- Blue cluster: Includes case study, forensic engineering, accident analysis, and root cause analysis—emphasizing empirical, post-event investigation practices.

Notably, the keyword “collapse” is the largest node, indicating its central role in the literature. It is strongly connected to “structural analysis”, “risk analysis”, and “progressive collapse”, reflecting the primary focus on understanding failure mechanisms through analytical and simulation-based approaches. The term “accident analysis” also occupies a central position, linking methodological frameworks with real-world events.

This co-occurrence network highlights the interdisciplinary nature of collapse investigations, where engineering modeling, risk assessment, and forensic methodologies converge. It also reveals the growing integration of data-driven and hybrid models, as seen in the connections between machine learning, Bayesian networks, and reliability analysis.

The analysis not only maps the current research landscape but also identifies emerging trends, such as the shift from isolated failure assessments to systemic risk modeling. These insights can guide future research in developing more comprehensive, integrated, and predictive frameworks for structural safety and disaster prevention.

3.5. Temporal Distribution of Included Publications

The 101 publications in this review show a clear upward trend in research on structural collapse and risk analysis over the last 15 years. As shown in Figure 8, the number of papers per year was very low in the early 2010s, with only 1 to 4 publications per year between 2010 and 2014. This initial phase was the infancy of systematic accident analysis in civil engineering, often limited to case specific criminal reports.

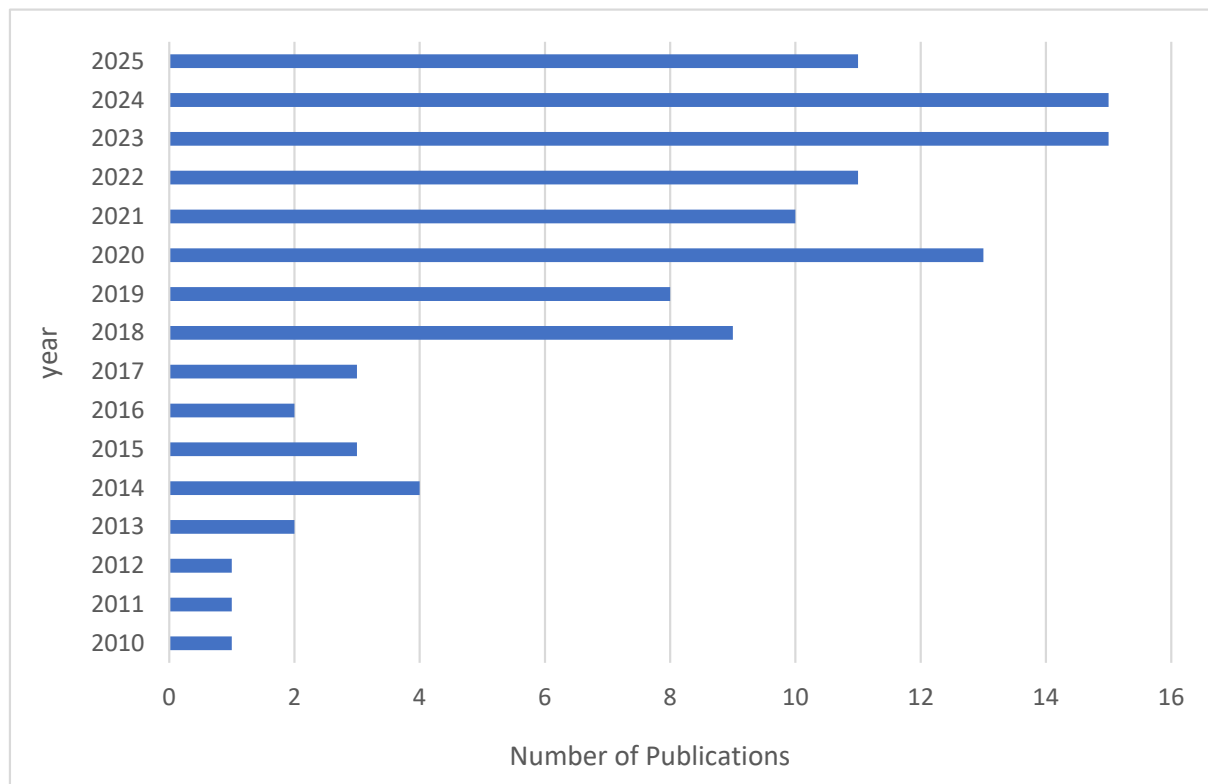


Figure 8. Temporal distribution of included publications.

There was a moderate increase between 2015 and 2019, with 3 publications in 2015, 9 in 2018 and 8 in 2019. This period coincided with high profile disasters such as the Morandi

Bridge collapse (2018) and the Brumadinho dam disaster (2019) which likely triggered academic interest in failure mechanisms and risk modeling.

But the most growth started in 2020 with 13 publications—the highest number of papers at that time. This growth continued in 2021 (10) and 2022 (11) and peaked in 2023 and 2024 with 15 publications each year. The high output in these years indicates a mature research field where risk analysis is no longer an ad hoc response to disasters but an established area of research.

Even in 2025 with 11 publications already documented, the field is not slowing down, so structural safety and accident modeling have become a permanent priority in civil engineering research.

This timeline shows a shift from isolated and reactive studies to a more systematic and proactive approach to understanding structural failures. The growing number of papers supports the development of standardized risk assessment frameworks, hybrid modeling techniques and data-driven methodologies that can be applied to different types of structures and hazards.

3.6. Evolution and Integration of Risk Analysis Models

The field of risk analysis in structural collapse investigations has undergone significant methodological evolution between 2010 and 2025. While earlier studies relied on isolated, single-method approaches, recent research increasingly adopts integrated and hybrid frameworks that combine complementary strengths to address the complexity of real-world failures. This section traces the historical development and integration pathways of key analytical models, highlighting how foundational frameworks have been adapted, extended, and combined over time.

A visual representation of these evolutionary and integrative pathways is presented in Figure 9, which maps the chronological development, adaptation, and hybridization of key risk analysis models used in structural collapse investigations from 2010 to 2025.

- From Fault Trees to Hybrid Risk Frameworks

The Fault Tree Analysis (FTA) model, introduced in 1962, has served as a foundational tool for causal reasoning in accident investigations. Starting in 2013, researchers began transforming FTA into Bayesian networks (BN) to better handle uncertainty and probabilistic dependencies [20]. This shift marked a move from binary logic to probabilistic inference, enhancing the model's applicability in complex systems. By 2017, FTA was combined with the Analytic Hierarchy Process (AHP) to support multi-criteria decision-making under risk [30]. The trend continued into 2023 and 2024, with studies integrating FTA, AHP, and BN into unified frameworks [18,25], reflecting a growing emphasis on systemic risk modeling that accounts for both technical and organizational factors.

- Computational Models: Advancing from FEA to Multi-Physics Simulation

Finite Element Analysis (FEA), originating in the 1940s, has remained a cornerstone of structural simulation. Between 2013 and 2022, its applications evolved from nonlinear static analysis to dynamic 3D modeling using advanced software such as OpenSees, ABAQUS, and LS-DYNA [35,58,66]. A notable shift occurred in 2025 with the adoption of a combined CFD-FE (Computational Fluid Dynamics–Finite Element) model, marking the integration of fluid-structure interaction in collapse analysis—particularly relevant for fire- or wind-induced failures [22]. This evolution reflects a broader trend toward multi-physics simulation, where mechanical, thermal, and aerodynamic effects are analyzed simultaneously.

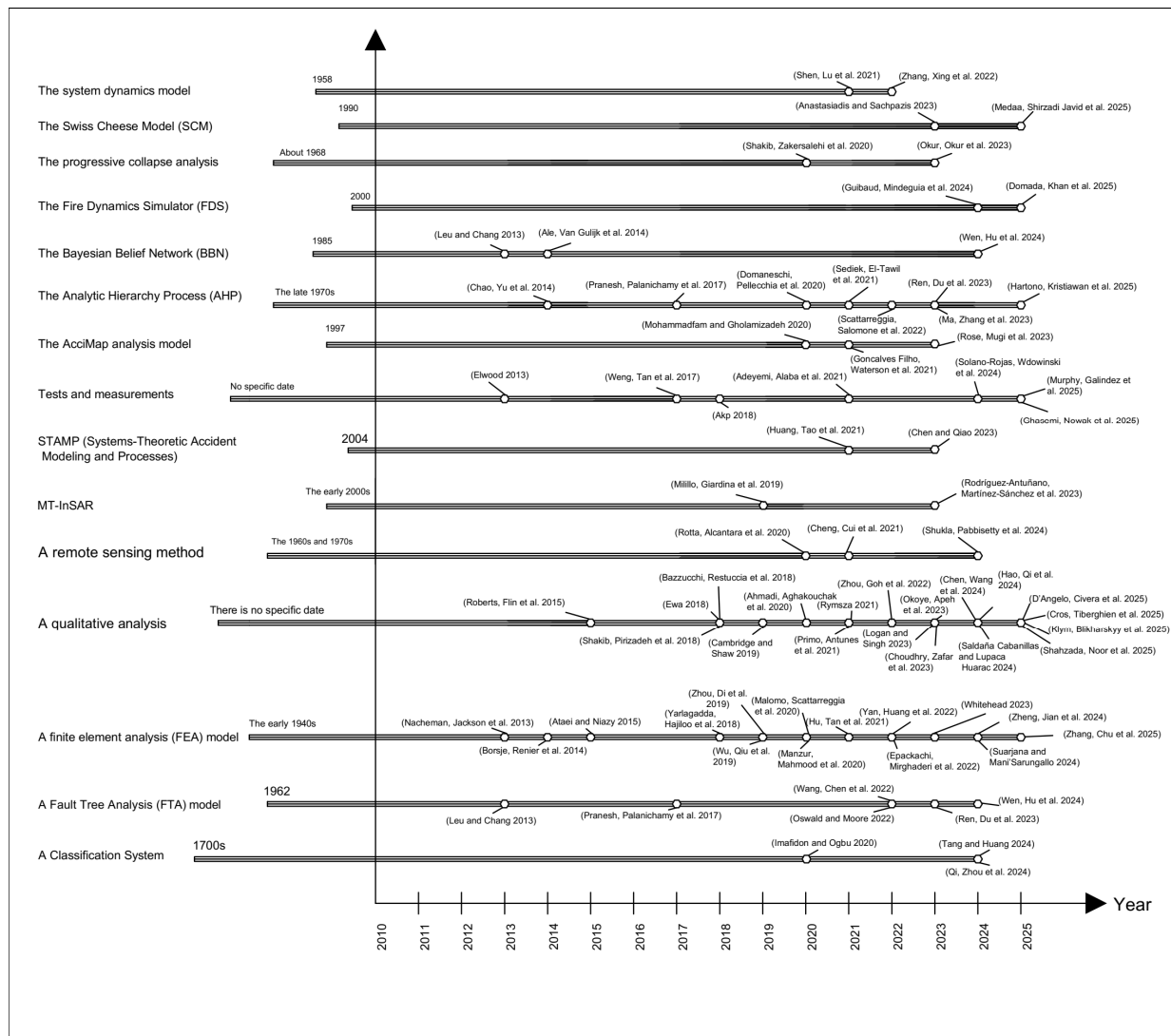


Figure 9. Evolution and integration pathways of risk analysis models (2010–2025).

- **Qualitative and Systemic Approaches: From Case Studies to Integrated Diagnostics**

Qualitative methods have also evolved beyond descriptive case studies. The AcciMap model (1997) reappeared in 2020 and 2021 with enhancements that link organizational levels to failure events [3,62]. Similarly, STAMP (Systems-Theoretic Accident Modeling and Processes, 2004) was extended in 2023 into a hybrid framework combining fuzzy DEMATEL and ISM, enabling deeper analysis of socio-technical interactions [75]. These developments show a clear shift from post-event description to systemic diagnosis, where human, organizational, and technical factors are analyzed as interconnected layers.

- **Emergence of Data-Driven and Geospatial Integration**

Remote sensing and geospatial methods, rooted in the 1960s–1970s, have seen renewed application since 2019. The use of MT-InSAR (Multi-Temporal InSAR) for monitoring infrastructure deformation has become more sophisticated, integrating time-series analysis and probabilistic modeling (e.g., MCMC) [42]. By 2024, these methods were combined with GIS to create multi-hazard risk assessment models, enabling large-scale, post-event damage evaluation [95]. This reflects a growing reliance on Earth Observation (EO) data for forensic engineering, especially in dam and urban infrastructure failures [80].

- **Integration of Testing, Measurement, and Reliability Models**

Empirical and measurement-based models have also advanced. The displacement-based framework (2013) [4] and material-based risk models (2018) [34] have been integrated with field inspections, construction documentation review, and material testing—culminating in 2025 with a system-level and element-level reliability model [12]. This holistic approach, applied in investigations like the Francis Scott Key Bridge collapse, highlights a shift toward evidence-based validation of analytical results [12].

- **Synthesis: The Rise of Hybrid and Adaptive Frameworks**

The most significant trend across all domains is the integration of models across methodological boundaries. Examples include:

- FEA + CFD for fire-induced collapse.
- FTA + AHP + BN for probabilistic risk assessment.
- Remote sensing + GIS + statistical modeling for post-disaster assessment.
- LLMs + qualitative analysis for extracting insights from unstructured data (e.g., news reports) [90].

These hybrid approaches reflect a response to the increasing complexity of structural systems and the limitations of single-method analyses. They also indicate a maturation of the field, where risk analysis is no longer a standalone tool but a dynamic, adaptive process that combines computational power, empirical data, and systemic thinking.

4. Systematic Literature Review & Critical Analysis

This section presents a critical synthesis of the risk analysis models used in the 101 included studies, moving beyond mere description to evaluate their methodological strengths, limitations, and contextual applicability. Unlike the scientometric and evolutionary analyses in previous sections, this part focuses on critical assessment—examining not just what models were used, but how effectively they were applied, what insights they provided, and where they fell short in explaining structural collapse mechanisms.

The analysis is organized thematically around major modeling approaches, with each subsection dedicated to a specific category of risk analysis. For each model, we evaluate its performance based on the authors' own reported advantages and limitations (as cited in the original studies).

Special attention is given to the interplay between model type, disaster context, and data availability, highlighting how certain frameworks excel in specific scenarios (e.g., FEA in structural simulation, FTA in causal mapping, or qualitative models in organizational analysis). The goal is not to rank models hierarchically, but to understand their fitness for purpose and identify patterns in their success or failure.

By systematically comparing and contrasting these approaches, this section lays the groundwork for the broader discussion in Section 5, where we explore the implications for future research, practice, and integrated risk assessment frameworks.

4.1. Finite Element Analysis (FEA) and Computational Models

Finite Element Analysis (FEA) is the most widely used computational framework for studying structural collapses, appearing in 15 studies across the review period (2010–2025). Its popularity stems from its ability to simulate complex structural behaviors under extreme loading conditions, providing detailed information on stress, deformation and failure progression. However, while FEA offers high fidelity in modeling, its effectiveness is often limited by data availability, modeling assumptions and computational complexity.

- **Capabilities and Strengths**

FEA's main strength is in detailed structural insight and quantitative simulation. In the Indiana State Fair stage collapse study, nonlinear FEA was used to analyze the structural

response and was validated against field data [26]. Similarly, in the Morandi Bridge collapse study, FEA was very reliable in identifying failure mechanisms when combined with monitoring data [44]. For the FIU pedestrian bridge, FEA models estimated the time-to-collapse and simulated real-world conditions, integrating forensic observations [41,58].

The model was also used in fire-induced collapse scenarios. In the Plasco Building case, a 3D FEA model (OpenSees) combined thermal analysis and simulated traveling fire, enabling progressive collapse simulation [35]. Later, an advanced LS-DYNA-based model was used to model fracture of steel, welds and concrete, capturing complex contact behaviors [66]. For the Mexico City Metro, a 3D-FDM model with FLAC 3D software predicted vertical settlement and provided design recommendations to prevent differential deformations [19].

Hybrid approaches have further increased FEA's applicability. The combined CFD-FE model used in the Arecibo Telescope study simulated wind loads at different speeds, helping to predict structural responses to hurricane forces [22]. Similarly, the integration of FDS and OpenSees in the Plasco Building fire study simulated fire dynamics, heat transfer and structural response [24].

- Limitations and Critical Challenges

Despite its strengths, FEA is often criticized for its complexity, assumptions and data dependency. Several studies show that FEA results are very sensitive to input data quality. For instance, the Morandi Bridge collapse study was limited by the need for monitoring data to obtain optimal results [44], the FIU bridge simulation required significant simplification of the failure process and couldn't pinpoint the initial cause [58].

With limited structural information, FEA's accuracy is compromised. The Plasco Tower modeling was limited by incomplete configuration data, many assumptions were made that affected the outcome [35,66]. The Rana Plaza study noted that FEA can be computationally intensive and complex, often requiring simplifications that reduce the results [7].

Moreover, some applications lack comprehensiveness. The nonlinear dynamic analysis of a self-anchored suspension bridge adopted a limited range of scenarios and environmental factors, reducing its generalizability [38]. The Tretten Bridge study used linear static modeling, which failed to account for nonlinear material behavior and could not be experimentally validated [83]. The FIU bridge analysis, while detailed, was confined to a narrow scope and did not address broader systemic causes [40].

- Contextual Performance and Model Evolution

FEA has proven particularly effective in post-event forensic simulation and progressive collapse assessment, as seen in the FIU bridge [48,58] and garment factory buildings in Bangladesh [7]. However, its predictive capability remains limited, as most applications are retrospective. The model excels when integrated with real design data and field observations [26,40], but struggles in scenarios with poor documentation or dynamic loading conditions.

The trend toward hybrid FEA models—such as CFD-FE for wind loads [22] and FDS-FEA for fire-structure interaction [24]—reflects a strategic evolution to address multi-hazard scenarios. These integrations enhance realism but also increase computational cost and complexity [24,66].

In summary, FEA remains a cornerstone of structural collapse analysis, valued for its detail, flexibility, and simulation power. However, its effectiveness is contingent on data quality, modeling expertise, and appropriate scope. Future applications should focus on improving data integration, reducing assumptions, and expanding predictive capabilities through real-time monitoring and machine learning-assisted calibration.

4.2. Fault Tree Analysis (FTA) and Hybrid Risk Frameworks

Fault Tree Analysis (FTA), introduced in the 1960s, remains a fundamental tool for causal reasoning in structural collapse investigations. Its top-down, structured logic allows for the systematic identification of failure paths, which is particularly valuable in complex systems where multiple factors converge. Over the last decade, FTA has evolved beyond its traditional deterministic form and is increasingly being combined with probabilistic and decision-making models to enhance its analytical depth. This section evaluates the application of FTA and its hybrid variants, highlighting its strengths in risk prioritization and limitations in addressing systemic and dynamic factors.

- Capabilities and Strengths

The main strength of FTA lies in its ability to identify failure causes and support both qualitative and quantitative analysis. In the Kutai Kartanegara Bridge collapse investigation, FTA mapped the sequence of events leading to suspender loss, providing a solid basis for preventive measures [63]. In the Thane building collapse study, FTA identified potential failure points in a structured way, enabling integration with other risk assessment methods [64].

Combining FTA with the Analytic Hierarchy Process (AHP) has improved its decision-making capabilities. For example, in the Deepwater Horizon disaster, FTA and AHP were used to quantify key human and technical factors, contributing to a deeper understanding of the accident's root causes [30]. In construction safety assessments, this hybrid approach is used to prioritize risks and propose targeted mitigation measures, such as improved material inspection and maintenance procedures [25].

Further advancement is seen in combining FTA with Bayesian Belief Networks (BBN), which transforms binary logic into probabilistic reasoning. The Bayesian-network-based safety risk assessment model for steel construction projects addressed the limitations of traditional FTA by calculating risk probabilities and generating early warnings [20]. In the post-earthquake assessment of subway stations, the FTA-BN combination provided a detailed, quantitative simulation of seismic damage, supporting emergency response planning and informed decision-making [18].

These hybrid models reflect a clear shift from descriptive to predictive and adaptive risk assessment, where uncertainties are explicitly modeled and probabilities are dynamically updated as new data becomes available.

- Limitations and Critical Challenges

Despite its strengths, FTA has limitations, especially when used in isolation. Several studies show that FTA alone may not provide a complete picture of the risk landscape. For example, the FTA model for the Kutai Kartanegara Bridge did not assess failure probabilities and required complementary methods for a full evaluation [63]. The standalone FTA used in the Thane building collapse study may not account for non-quantifiable risks or external factors outside the fault tree [64].

The major limitation is the overemphasis on technical and procedural factors, often at the expense of human and organizational dynamics. The Deepwater Horizon analysis, while identifying key human errors, did not systematically address the broader Human and Organizational Factors (HOFs) in accident causation [30]. This is a recurring issue: FTA treats human error as a terminal event rather than a symptom of deeper systemic flaws.

While hybrid models are more robust, they introduce new challenges. The FTA-AHP combination is effective for risk prioritization but can be complex and time-consuming, as both methods are often applied separately before integration [25]. Concerns about accuracy in traditional analytical methods further complicate its application [25].

Similarly, the FTA-BN integration is powerful but data-intensive and complex. The Bayesian network's accuracy depends heavily on the quality and quantity of data for training and investigation [20]. In the subway station assessment, the model required detailed seismic performance data for each component, limiting its applicability in data-scarce environments [18]. Combining methods increases analytical complexity and may hinder practical implementation [18].

- Contextual Performance and Model Evolution

FTA is most effective in engineering-centric investigations where failure sequences are well-defined, such as bridge collapses [63,64] and industrial accidents [30]. Its hybrid forms, especially FTA-AHP and FTA-BN, are well-suited for multi-criteria decision-making and probabilistic risk assessment in safety-critical environments like subway systems [18] and steel construction [20].

However, FTA is less effective in socio-technical systems where interactions are dynamic and non-linear. Its static, hierarchical structure is ill-suited to capturing feedback loops and emergent behaviors that characterize complex organizational failures.

The evolution from deterministic FTA to probabilistic hybrid models reflects a broader trend toward systemic risk modeling. Yet, as the SAMA model [97] and STAMP-based approaches [61] suggest, future frameworks must go beyond causal chains to model the synergy and accumulation of risks across technical, organizational, and environmental domains.

In summary, FTA remains a valuable and widely used tool for structuring accident investigations, particularly when enhanced with AHP or BBN. However, its effectiveness is limited by data dependency, complexity, and a narrow technical focus. To address modern structural failures, FTA must evolve further—toward models that are dynamic, systemic, and human-centric.

4.3. Qualitative, Forensic, and Systemic Approaches

While computational and probabilistic models dominate the technical analysis of structural failures, qualitative, forensic, and systemic approaches are essential for understanding the human, organizational, and contextual dimensions of disasters. These methods are particularly valuable when physical data is limited or when root causes extend beyond structural flaws into governance, ethics, and decision-making processes. This section evaluates the application of these non-computational frameworks, highlighting their strengths in holistic diagnosis and their limitations in generalizability and predictive power.

- Capabilities and Strengths

Qualitative methods excel in contextual understanding and multi-stakeholder integration. The forensic engineering investigation of the Plasco Building collapse [49] combined eyewitness accounts, visual evidence, and structural mechanics to provide a comprehensive, multi-source analysis. Similarly, the forensic analysis of the Morandi Bridge collapse [59] enabled a clear identification of the failure mechanism and a detailed categorization of causes, despite reliance on historical data.

The AcciMap model has proven effective in mapping organizational and technical failures across multiple levels. In the Plasco Building investigation, the Timed MTO and AcciMap methods allowed for a detailed categorization of causes and facilitated a deeper understanding of systemic interactions [3]. The Contributing Factors Classification System Framework (CFCFSF) based on AcciMap [62] provided a highly reliable and structured classification of accident causes, covering administrative, governmental, and technical factors. Its structured approach makes it a promising tool for standardized accident reporting.

Systemic models like STAMP (Systems-Theoretic Accident Modeling and Processes) treat accidents as outcomes of complex socio-technical systems rather than simple chains of events. The Enhanced STAMP model applied to the Xinjia Express Hotel collapse during the pandemic provided a structured framework for identifying root causes and classifying causal factors [61]. A hybrid approach combining STAMP with fuzzy DEMATEL and ISM further enhanced analytical depth by quantifying the strength of causal relationships [79], demonstrating the potential of integrating qualitative and semi-quantitative methods.

Emerging data-driven qualitative methods are also gaining traction. The use of large language models (LLMs) to extract and analyze data from news reports on urban ground collapses achieved high recall and created an open-access inventory [90]. Similarly, categorical data analysis of metro construction accidents helped identify high-risk scenarios and inform training decisions [70]. The sensemaking framework centered on “senselisting” offered a practical perspective for crisis management organizations, facilitating a shift from individual to collective action [100].

- Limitations and Critical Challenges

Despite their strengths, qualitative approaches face significant limitations, particularly in objectivity, data quality, and predictive capability. Many studies rely on secondary or unreliable sources, which can introduce bias. For example, the case study on the Champlain Towers South collapse [81] relied on media reports suspected of bias and lacked access to design documents, limiting its technical evaluation. Similarly, the multidisciplinary case study on the 2025 Bangkok skyscraper collapse [103] depended on unreliable data, affecting the validity of its analysis.

A recurring issue is the descriptive nature of these models. The qualitative geotechnical assessment of the Brumadinho dam disaster [43], while insightful, could not predict the failure and carried uncertainty in estimation. The qualitative, literature-based review on reinforced concrete bridge damage [102] lacked quantitative assessment and predictive capability. The statistical database approach for Italian bridge collapses [99] identified prevalent causes but did not provide detailed engineering analysis of failure mechanisms.

Furthermore, many qualitative models are context-specific and not easily generalizable. The sensemaking framework [100] was developed for a unique crisis and is not applicable in all contexts. The Enhanced STAMP model [61] was specific to the pandemic period, limiting its generalizability. The Driller’s Situation Awareness (DSA) model [2] was based on limited first-hand reports and provided only a preliminary analysis.

Complexity and resource intensity are also barriers. The hybrid STAMP-fuzzy DEMATEL-ISM approach [79] requires significant expertise and time-consuming data collection. The risk communication framework for mining disasters [56] focused on post-disaster communication failures but did not address structural engineering details, limiting its completeness.

- Contextual Performance and Model Evolution

Qualitative and forensic methods are most effective in post-event investigations where the goal is to assign responsibility, understand systemic failures, and propose preventive measures. They are particularly valuable in complex socio-technical systems such as dam collapses (Brumadinho, [21]), construction sector failures (Lagos, [53]), and high-profile building collapses (Plasco, [49]; Champlain Towers, [81]).

The integration of data science tools—such as LLMs [90] and categorical analysis [70]—marks a significant evolution, enhancing the scalability and objectivity of qualitative methods. Similarly, the combination of systemic models with semi-quantitative techniques (e.g., STAMP + DEMATEL, [79]) reflects a growing effort to bridge the gap between narrative-based and data-driven analysis.

However, a critical gap remains: the lack of predictive capability. Most qualitative models are retrospective, offering insights into “what happened” and “why,” but not “what might happen next.” To become more impactful, these approaches must evolve toward proactive, real-time monitoring frameworks that integrate qualitative insights with early warning systems.

In summary, qualitative, forensic, and systemic approaches provide indispensable insights into the human and organizational dimensions of structural failures. They complement computational models by addressing the “soft” factors that often underlie catastrophic events. However, their effectiveness is constrained by data quality, subjectivity, and limited predictive power. Future research should focus on hybridizing these methods with real-time data streams and machine learning to create more dynamic, adaptive, and actionable risk assessment frameworks.

4.4. Measurement-Based, Testing, and Reliability Models

Measurement-based, testing, and reliability models are crucial for grounding risk analysis in real-world data and empirical evidence. Unlike computational or qualitative frameworks, these approaches rely on field observations, material testing, structural monitoring, and statistical validation, making them particularly valuable for post-event investigation and forensic validation. This section evaluates their application in structural collapse studies, highlighting their strengths in objectivity and practical relevance, as well as their limitations in scalability and predictive capability.

- **Capabilities and Strengths**

The primary strength of these models lies in their direct connection to physical reality. The Structural Health Monitoring (SHM) framework was used in the Polcevera Viaduct collapse [50] to provide a proactive, data-driven approach to disaster prevention. By continuously monitoring structural behavior, SHM enables early detection of degradation and provides quantitative forecasts to support preventive maintenance [50]. In the Mexico City Metro collapse, a combination of satellite radar interferometry, leveling surveys, and subsurface profiles allowed for high-accuracy prediction of future geohazards [92].

The displacement-based framework has proven effective in seismic performance evaluation, particularly in the Christchurch earthquake studies [4]. Its applicability in real-world scenarios makes it a practical tool for assessing building safety under dynamic loads. In the FIU pedestrian bridge case, a system-level and element-level reliability model was used to identify structural weaknesses and assess target reliability, contributing to a comprehensive understanding of the collapse mechanism [12].

Field-based empirical analyses also provide valuable insights. A study on mechanical services in Lagos buildings used descriptive statistics, Chi-square tests, and correlation analysis to identify specific risk factors contributing to instability [57]. This data-driven approach enhances the credibility of findings and supports targeted mitigation strategies.

Moreover, in situ inspections and material testing have been integral to forensic investigations. In the Chirajara bridge collapse, a combination of global analyses, refined nonlinear modeling, construction documentation review, and material testing enabled a comprehensive evaluation of the failure mode [98]. This multi-source approach ensures that conclusions are not based on assumptions, but on verifiable, physical evidence.

- **Limitations and Critical Challenges**

Despite being evidence-based, these models face significant challenges, particularly in cost, complexity, and generalizability. The SHM system used in the Morandi Bridge investigation, while powerful, required significant investment and was conducted post-collapse, meaning it was not used for real-time early warning [50]. Similarly, the multi-

method approach in the Mexico City Metro study [92] is complex and resource-intensive, limiting its applicability in low-budget or data-scarce environments.

Another limitation is data quality and availability. The displacement-based framework, while practical, has limited predictive accuracy and requires careful calibration [4]. The system-level reliability model for the Francis Scott Key Bridge [12] does not account for structural redundancy, which may invalidate its results in certain scenarios.

Furthermore, some models are descriptive rather than predictive. The field-based empirical analysis in Lagos [57] provides data-driven insights but lacks system-level modeling of risk interactions, limiting its ability to forecast cascading failures. The two-scale numerical model used in the Miami Dade College parking garage collapse [31] requires complex calibration and incremental sampling, making it difficult to apply broadly.

- Contextual Performance and Model Evolution

These models are most effective in post-event forensic investigations and preventive monitoring of critical infrastructure. They perform best when integrated with design data, inspection reports, and historical records—as seen in the FIU bridge [58], Chirajara bridge [98], and Polcevera Viaduct [52] studies.

A notable trend is the integration of multiple data sources—such as satellite observations, leveling surveys, and structural parameters—to enhance accuracy and predictive power [92]. This multi-method approach reflects a shift toward hybrid, evidence-based frameworks that combine measurement with simulation.

However, a key gap remains: real-time implementation. Most applications are retrospective or require extensive setup. To increase impact, future research should focus on automating data collection, reducing costs, and developing standardized protocols for reliability assessment.

In summary, measurement-based, testing, and reliability models provide robust, evidence-based insights into structural failures, making them indispensable for forensic analysis and infrastructure management. Their strength lies in objectivity and verifiability, but their effectiveness is constrained by cost, complexity, and data dependency. Future advancements should prioritize scalability, integration with digital twins, and real-time monitoring systems to enhance their preventive potential.

4.5. Remote Sensing, Geospatial, and Data-Driven Methods

In recent years, remote sensing, geospatial analysis, and data-driven methods have become powerful tools for monitoring structural integrity, assessing post-event damage, and identifying early warning signs of collapse. These approaches leverage satellite imagery, Earth Observation (EO) data, and advanced computational techniques to provide large-scale, non-invasive insights into infrastructure health. This section evaluates their application in structural collapse investigations, highlighting their strengths in spatial coverage and predictive monitoring, and their limitations in predictive accuracy, data dependency, and real-time implementation.

- Capabilities and Strengths

Remote sensing and geospatial methods excel in large-scale monitoring and pre-collapse detection. The multi-method remote sensing and geospatial analysis framework applied to the Brumadinho dam disaster [21] provided comprehensive spatial coverage and enabled time-series analysis of conditions in the months leading up to the collapse. Similarly, the multi-geometry InSAR (Interferometric Synthetic Aperture Radar) analysis with MCMC used in the Morandi Bridge case [42] achieved millimetric accuracy in characterizing three-dimensional displacements, offering potential for early warning systems.

The integration of Earth Observation (EO) and Multi-Temporal InSAR (MT-InSAR) has further enhanced monitoring capabilities. A methodology for assessing urban infrastructure vulnerability in Mexico City demonstrated the ability to detect early signs of structural instability and support continuous monitoring [80]. This enables proactive risk management by identifying subsidence, deformation, and ground movement before catastrophic failure occurs.

For post-event assessment, remote sensing proves highly effective. The post-event damage assessment methodology based on remote sensing and geospatial analysis used after the Francis Scott Key Bridge collapse [95] enabled rapid, emergency-level damage analysis under all conditions, contributing to emergency response planning. Its ability to provide measurable, comprehensive results makes it a valuable tool for disaster recovery operations.

Data-driven methods are also gaining traction. The use of large language models (LLMs) to extract and analyze urban ground collapse events from news reports achieved high recall and created an open-access inventory [90]. Similarly, categorical data analysis of metro construction accidents helped identify high-risk scenarios and inform training decisions [70]. These approaches enhance scalability and objectivity by transforming unstructured data into actionable insights.

- Limitations and Critical Challenges

Despite their strengths, these methods face significant challenges, particularly in predictive power, data quality, and operational complexity. A recurring issue is that most applications are retrospective, conducted after the collapse, which limits their preventive value. For example, the InSAR analysis of the Morandi Bridge [42] was performed post-collapse and was not used for real-time early warning, despite its potential.

Another limitation is data dependency and accuracy. The MT-InSAR methodology in Mexico City [80] requires extensive data processing and validation, and its reliability depends on the quality of satellite imagery. The LLM-based inventory construction [90] achieved high recall but low precision (below 35%), resulting in a high rate of false positives that require manual verification.

Furthermore, many models lack predictive capability. The geospatial risk assessment framework for the Mecca crane collapse [94] does not predict the likelihood or severity of collapse and relies on expert opinion, which may introduce bias. The remote sensing system used for Brumadinho [21] failed to predict the future and could not reveal the scenario preceding the collapse, despite its detailed spatial coverage.

Complexity and resource intensity are also barriers. The hybrid STAMP-fuzzy DEMATEL-ISM approach [79], while powerful, requires significant expertise and time-consuming data collection. Similarly, the integrated FDS-OpenSEES simulation for the Plasco Building fire [24] is computationally intensive and requires detailed building configuration data, limiting its applicability in data-scarce environments.

- Contextual Performance and Model Evolution

These methods are most effective in large-scale infrastructure monitoring (e.g., dams, bridges, urban systems) and post-disaster damage assessment. They perform best when integrated with in situ inspections, structural data, and historical records—as seen in the Mexico City Metro [92], Brumadinho [21], and Morandi Bridge [42] studies.

A notable trend is the integration of multiple data sources—such as satellite observations, leveling surveys, and structural parameters—to enhance accuracy and predictive power [92]. This multi-method approach reflects a shift toward hybrid, evidence-based frameworks that combine remote sensing with simulation and field data.

However, a key gap remains: real-time, proactive implementation. Most applications are retrospective or require extensive setup. To increase impact, future research should focus on automating data processing, improving model precision, and developing standardized protocols for early warning systems.

In summary, remote sensing, geospatial, and data-driven methods provide scalable, non-invasive insights into structural safety, making them invaluable for infrastructure monitoring and disaster response. Their strength lies in spatial coverage and early detection, but their effectiveness is constrained by data quality, complexity, and limited predictive power. Future advancements should prioritize real-time integration, improved accuracy, and accessibility to transform these tools into proactive, decision-support systems for structural safety.

4.6. Single-Use and Context-Specific Models: Innovation, Isolation, and Untapped Potential

Alongside the widely adopted and evolving frameworks discussed in previous sections, a diverse set of risk analysis models has been applied only once across the 101 included studies. These single-use models, while not part of a broader methodological trend, reflect the adaptability, innovation, and contextual specificity of analytical approaches in structural collapse investigations. A comprehensive summary of these models—including their advantages, limitations, and reference numbers—is provided in Table 3.

Table 3. Categories and topics identified.

Risk Analysis Model	Year	Key Advantages	Key Limitations	Reference
Blind Zones & Vicious Cycles	2010	Structured critique of risk management foundations; highlights overlooked areas and ineffective cycles	May not lead to actionable improvements; descriptive rather than predictive	[1]
Linear and Nonlinear Analysis (ASCE/SEI)	2011	Standardized seismic evaluation; identifies vulnerability indicators	Failed to predict specific collapse mechanisms; not suitable for complex buildings	[5]
Reliability-Based Risk Analysis	2015	Effective for analyzing safety under specific factors (e.g., wind); expandable	High uncertainty in predictions	[11]
HYRISK Model	2016	Detailed risk factor assessment; structured approach; integrable with management systems	Lacks robustness; assumptions do not always align with real data; validation issues	[8]
Multicriteria Analysis (Idrisi Selva®)	2018	Flexible; handles complex multi-criteria decisions	No reported limitations	[32]
Monte Carlo Simulation Model	2018	Comprehensive probabilistic analysis; simulates diverse contexts	Uncertainty due to random sampling variability	[33]
Joint Structure-Foundation-Soil Model	2019	Detailed understanding of soil-structure interaction; validated with real data	Resource-intensive; relies on numerous assumptions	[14]
RAM (IRD & IDF Indices)	2019	Systematic prioritization; considers economic impact	Requires unknown durability data; faces uncertainty	[39]
Improved Structural Vulnerability Theory (ISVT)	2019	Identifies weaknesses; quantitative; considers unforeseen events	Complex; depends on accurate data	[41]
Instantaneous Stiffness Degradation Method	2020	Effective for simulating bridge dynamic responses; simple to apply	May oversimplify real-world complexities; accuracy depends on multiple factors	[45]
Probabilistic and Modular Approach	2020	Comprehensive; tailored to specific building contexts	High uncertainty; complex to implement	[46]

Table 3. Cont.

Risk Analysis Model	Year	Key Advantages	Key Limitations	Reference
FMEA Model	2020	Systematic identification of failure modes; supports prioritization (RPN)	Subjective results; ignores component interactions; distorted risk rankings	[47]
Structural Health Monitoring (SHM)	2020	Proactive; prevents disasters; provides quantitative forecasts	Requires significant investment	[50]
Capacity-Demand Time-Domain Model	2020	Practical; high predictive ability; uses historical data	Lacks real-time monitoring; relies on archival data	[51]
Computational Simulation Model	2021	Detailed understanding of collapse mechanisms; supports performance-based design	Tailored to specific scenarios (e.g., overheight impacts); lacks real-world validation	[10]
Sendai Framework for DRR	2021	Universally recognized; promotes prevention and “Build Back Better”	Lacks binding mechanisms; shifted toward promotion over prevention	[56]
CFD Fire Modeling	2022	Realistic fire spread simulation; integrable with structural analysis	Requires extensive calibration; data-limited in post-disaster scenarios	[65]
Accident Mechanism of Multiple Processes (AMMP)	2022	Captures multi-process accidents; considers space and time; identifies safety measures	Not generalized; not predictive; complex	[67]
Probabilistic Risk Assessment (PRA)	2023	Comprehensive risk estimation; supports cost-benefit analysis of design measures	Limited applicability in regions with low threat likelihood; progressive collapse is rare in Western contexts	[72]
Computational Fluid Dynamics (CFD) Model	2023	Simulates complex environmental interactions; supports scenario-based recommendations	Limited directional analysis of wind effects	[74]
FITB (Fire Investigation for Tall Buildings)	2023	Realistic fire behavior simulation; understands complex fire dynamics	Limited visual evidence; requires simplifications; hard to model vertical spread	[77]
Life Safety Model (LSM)	2023	Simulates flood impacts; estimates loss of life and evacuation times	Sensitive to parameters; complex; data-intensive	[78]
QBDC-Based Active Learning + DL	2024	High computational efficiency; <0.5% error in reliability calculation	Traditional methods are expensive; new approach needs further validation	[88]
Comparative Machine Learning Modeling	2024	Predictive power; identifies key risk factors via feature importance	Sensitive to noise; high computational cost; data-dependent	[15]
Consequence-Based Robustness Model	2025	Analyzes alternative load paths; supports design; detailed response analysis	Computationally complex; data-intensive; scenario-specific	[101]

The HYRISK model [8] and the reliability-based risk analysis [11] were developed for targeted assessments of bridge safety, offering structured yet isolated contributions to risk evaluation. The joint structure-foundation-soil numerical model [14], applied to the Space Building collapse, provided a holistic view of geotechnical-structural interaction, though its resource-intensive nature limits broad applicability. Similarly, the Failure Mode and Effect Analysis (FMEA) model [47] and the RAM model with IRD and IDF indices [39] introduced systematic yet context-bound frameworks for prioritizing interventions and assessing degradation.

Emerging computational and data-driven methods also appear as one-time applications. The comparative machine learning modeling approach [15] demonstrated predictive power in identifying risk factors in Lagos building collapses, while the improved QBDC-based active learning with Deep Learning [88] showed high efficiency in dynamic reliability analysis. These reflect a growing interest in AI-assisted risk forecasting, even if not yet standardized.

Other models are deeply tied to specific contexts:

- The Sendai Framework for Disaster Risk Reduction [56] was used as a policy-oriented lens for mining disasters.
- The Life Safety Model (LSM, [78]) and FITB (Fire Investigation for Tall Buildings, [77]) were developed for high-rise fire and flood safety.
- The consequence-based robustness assessment model [101] introduced a new perspective on post-collapse structural evaluation.

While these models lack widespread adoption, they demonstrate methodological innovation and the adaptation of tools to specific structural types, hazards, or regulatory environments. Their isolated use may stem from limited data availability, proprietary software, or the uniqueness of the investigated disaster. However, they represent valuable starting points for future research, particularly in niche areas such as tall building fires, soil-structure interaction, or AI-driven risk forecasting.

4.7. Classification Criteria for Risk Analysis Models

The classification criteria below are based on a critical synthesis of the advantages, limitations, and implementation contexts in Table 3 which provides an overview of the risk analysis models reviewed. To enable a systematic and transparent comparison of the risk analysis models in this review, a multi-dimensional classification framework was developed based on three key dimensions: cost, complexity, and data needs. This classification is not arbitrary but grounded in a rigorous content analysis of the advantages, limitations and implementation challenges reported across the 101 included studies. The criteria for each dimension are defined below to ensure consistency, reproducibility and practical relevance.

4.7.1. Cost Classification

The cost of a model refers to the overall resource intensity required for its application, including financial investment, computational demands and expertise. Models were categorized as:

- **High Cost:** Those that require substantial investment in proprietary software (e.g., ABAQUS, LS-DYNA), high-performance computing infrastructure or highly specialized personnel. Examples include Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD) and Machine Learning (ML) models [7,22,66]. These are often inaccessible in low-resource settings due to technical and financial barriers.
- **Medium Cost:** Models that require moderate resources, such as access to remote sensing data (e.g., MT-InSAR), sensor networks (e.g., SHM) or expert interpretation for hybrid frameworks (e.g., STAMP-fuzzy DEMATEL) [19,79]. While offering significant analytical value, they depend on existing data infrastructure and trained practitioners.
- **Low Cost:** Models that are simple to implement, rely on widely available tools or qualitative methods and have minimal computational or financial requirements. Examples include Fault Tree Analysis (FTA), Analytic Hierarchy Process (AHP) and the Swiss Cheese Model (SCM) [25,63,84]. These are particularly useful for rapid assessments and training in resource-constrained environments.

4.7.2. Complexity Classification

The complexity of a model reflects the complexity of the methodology, the number of interacting components and the level of technical expertise required for implementation. Models were classified as:

- **High Complexity:** Models with complex algorithms, multi-stage calibration or integration of multiple simulation domains (e.g., FDS-OpenSEES, ML-based predictive models) [15,24]. These often involve many assumptions and are sensitive to input quality so are hard to replicate without deep domain knowledge.
- **Medium Complexity:** Approaches that combine two or more analytical methods but are accessible to trained professionals (e.g., Bayesian Belief Networks, SAMA, FTA-BN) [18,91,97]. They offer a balance between depth and usability but require careful interpretation.
- **Low Complexity:** Straightforward, rule-based or descriptive models for rapid screening or initial investigation (e.g., classification systems, simple case studies) [13,53]. While easy to apply, they may oversimplify complex socio-technical realities.

4.7.3. Data Needs

The third dimension, data needs, was assessed based on type, volume, quality and accessibility of data required for the model to work. This dimension highlights the trade-off between predictive power and practicality.

- **High Data Needs:** Models that require large volumes of high resolution, time-series or real-time data, usually from sensors, remote sensing platforms or detailed structural records (e.g., SHM, MT-InSAR, machine learning) [42,80]. Performance is very sensitive to data gaps or inaccuracies.
- **Medium Data Needs:** Models using structured data such as inspection reports, design documents or historical incident databases (e.g., BBN, hybrid STAMP-fuzzy DEMATEL) [27,79]. They do not require continuous monitoring but rely on consistent and well documented inputs.
- **Low Data Needs:** Models that work with minimal or qualitative data, such as expert judgment, checklists or publicly available information (e.g., FTA, AHP, SCM, SAMA) [62,63,97]. These are useful in early-stage investigations or where comprehensive data is not available.

This three-dimensional classification system provides a solid foundation to analyze the applicability, strengths and limitations of different risk analysis approaches. It will be the basis for the strategic decision-support framework presented later in this discussion, so the final model is empirically grounded and methodologically sound.

5. Discussion

This review has synthesized 101 studies on risk analysis models used in structural collapse investigations, revealing a field in dynamic evolution. The findings are best understood through Figure 10, which presents a strategic framework for evaluating risk analysis models based on three key dimensions: complexity, data needs, and cost. This approach moves beyond classification to provide a practical decision support tool for selecting the right model for real applications.

The analysis shows that the most commonly used models such as Finite Element Analysis (FEA) and Fault Tree Analysis (FTA) are not inherently superior, but contextually optimal. Their effectiveness depends on the balance between complexity, data availability, and cost. As illustrated in Figure 10, this balance defines four distinct quadrants, each representing a different class of analytical approaches.

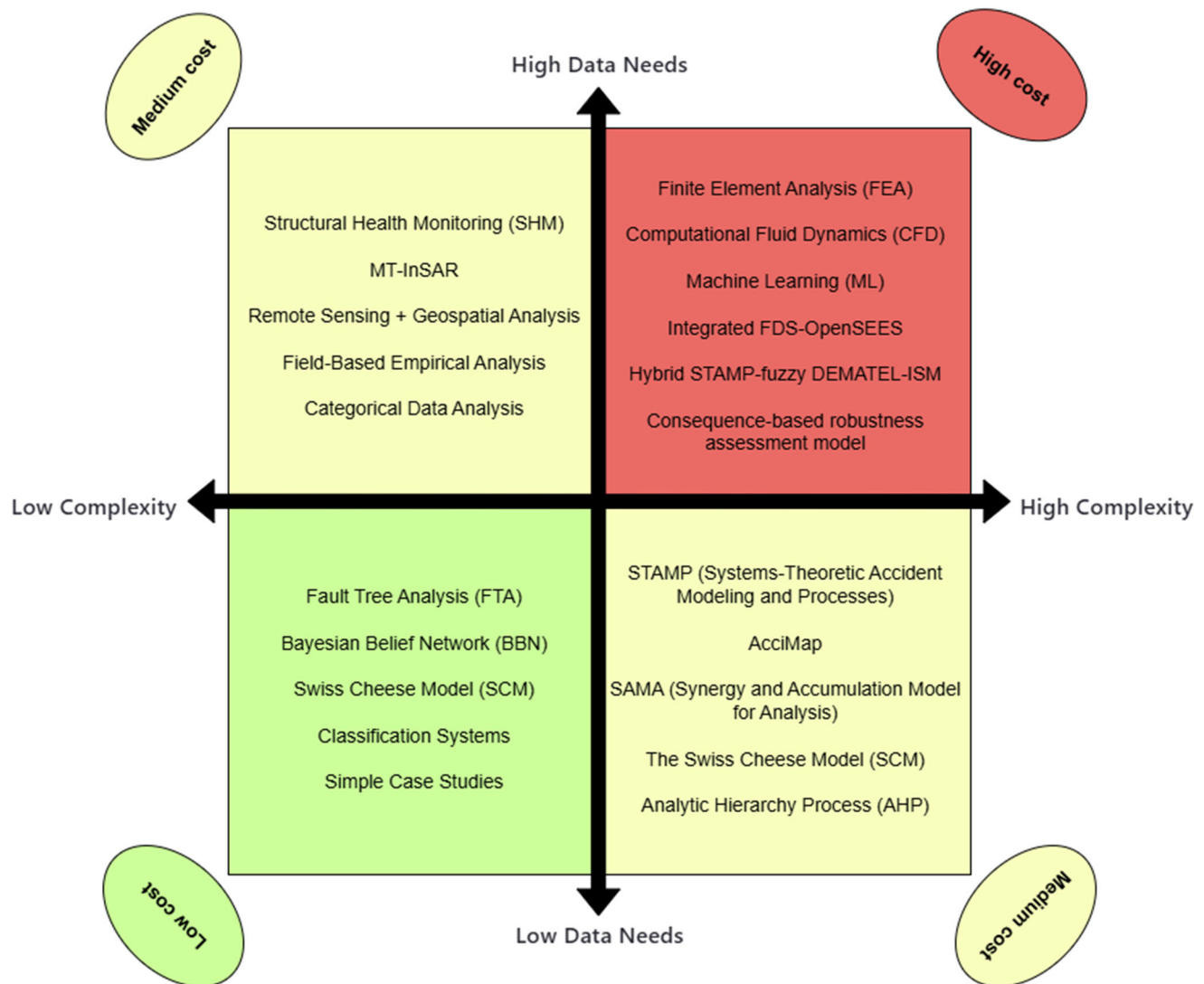


Figure 10. Strategic Framework for Selecting Risk Analysis Models in Structural Collapse Investigations.

5.1. Quadrant 1: Advanced Computational Models with High Complexity and High Data Needs (High Cost)

This quadrant contains the most sophisticated and accurate models, including FEA, Computational Fluid Dynamics (CFD), Machine Learning (ML), and integrated FDS-OpenSEES simulations. These models offer high fidelity and predictive power but are computationally intensive, data-hungry, and expensive. They are best suited for high-stakes, data-rich environments, such as forensic investigations of major disasters like the Plasco Building or Morandi Bridge collapses.

For example, the integrated FDS-OpenSEES model used in the Plasco fire investigation [24] combined fire dynamics with structural response, enabling a detailed, sequential simulation of the collapse mechanism. However, this model required detailed building configuration data and significant processing power, making it impractical for routine use. Similarly, machine learning models [15] demonstrated high accuracy in predicting building collapse risks but were limited by data quality and computational cost.

The high cost of these models is reflected in their resource requirements: specialized software, expert personnel, and extensive data collection. While they provide deep insights, their limited accessibility raises concerns about equity in structural safety.

5.2. Quadrant 2: Systemic and Qualitative Models with High Complexity and Low Data Needs (Medium Cost)

This quadrant includes systemic and qualitative models such as STAMP, AcciMap, AHP, and the Synergy and Accumulation Model for Analysis (SAMA). These models are complex in methodology but require minimal data, relying instead on expert judgment, case studies, and organizational analysis. They are ideal for post-event investigations where physical data is scarce but human and organizational factors are critical.

For instance, the STAMP model applied to the Xinjia Express Hotel collapse [61] treated the accident as a complex socio-technical system, identifying root causes across multiple levels. However, its qualitative nature limits its ability to predict future risks. Similarly, AcciMap [3] provided a comprehensive view of systemic failures in the Plasco Building disaster but could not quantify probabilities.

Among these models, the Synergy and Accumulation Model for Analysis (SAMA) [97] stands out as a novel and highly effective approach. Developed to address the limitations of traditional models, SAMA combines simplicity, low complexity, and minimal data requirements with a powerful focus on historical data and causal layering. Unlike models that rely on real time monitoring or complex simulations, SAMA leverages past failures as a foundation for future prevention, making it both accessible and deeply insightful.

These models are medium cost because they require skilled analysts but not expensive software or data infrastructure. Their strength lies in holistic understanding, but their lack of predictive capability remains a limitation.

5.3. Quadrant 3: Based and Remote Sensing Models with Low Complexity and High Data Needs (Medium Cost)

This quadrant includes measurement-based and remote sensing models such as Structural Health Monitoring (SHM), MT-InSAR, and remote sensing with geospatial analysis. These models are simple in design but demand large volumes of high-quality data. They are ideal for pre-collapse monitoring and emergency damage assessment, particularly in urban infrastructure.

For example, SHM systems used in the Polcevera Viaduct investigation [50] enabled continuous monitoring of structural behavior, providing proactive warnings. Similarly, MT-InSAR was used to detect millimetric displacements in the Morandi Bridge [42], offering early signs of instability. However, these models are not cost-effective without existing monitoring infrastructure. Their medium cost reflects the need for sensors, satellites, and data processing tools.

5.4. Quadrant 4: Simple and Descriptive Models with Low Complexity and Low Data Needs (Low Cost)

This quadrant contains the most accessible models, including FTA, Bayesian Belief Networks (BBN), Swiss Cheese Model (SCM), and simple case studies. These models are easy to apply, require minimal data, and are low-cost, making them ideal for rapid assessments, training, and low-resource settings.

For instance, FTA was used to analyze the Kutai Kartanegara Bridge collapse [63], systematically mapping failure pathways. Likewise, SCM was applied to the FC Twente stadium roof collapse [84], visualizing failures as layers of defense. However, these models are descriptive rather than predictive and may oversimplify complex realities. Their low cost makes them widely applicable, but their limited depth means they are best used as initial screening tools rather than comprehensive analyses.

In summary, the choice of a risk analysis model is not a question of superiority, but rather of fitness for purpose. It depends on the sensitivity of the facility being analyzed and studied, and whether the purpose of the study is a post-facto analysis or an analysis

of the facility's severity and vulnerability to collapse due to specific hazards. Figure 10 provides a powerful framework for navigating this decision space, showing that no single model is universally optimal. Instead, the most effective approach is strategic integration, combining models from different quadrants to address the full spectrum of failure causes.

An important observation from this review is that none of the 101 studies provided their risk analysis models as ready-to-use software tools, web applications, or open-source code packages. While many employed powerful simulation platforms (e.g., OpenSeES, ABAQUS, FDS), the models were implemented on a case-by-case basis and not released for broader reuse. This represents a critical gap in knowledge translation, as it limits the practical adoption of advanced methodologies by practitioners who lack specialized training or computational resources. Future research should aim not only to develop novel models but also to package them in accessible formats—such as Python libraries, cloud-based simulators, or plugins for common engineering software—to bridge the divide between academic innovation and field implementation.

The future of structural safety lies in developing hybrid models that balance accuracy, cost, and accessibility. For example, combining SHM (Quadrant 3) with FTA (Quadrant 4) can provide both real-time monitoring and causal reasoning at a manageable cost. Similarly, integrating ML (Quadrant 1) with AcciMap (Quadrant 2) can achieve predictive power.

The categorization of models in Figure 10—into low, medium, and high cost and complexity—is not arbitrary but grounded in the documented advantages, limitations, and implementation challenges reported across the 101 reviewed studies. For cost, models were classified as high if they required significant investment in software, hardware, or specialized personnel; medium if they demanded moderate resources and expert interpretation; and low if they were simple, widely accessible, and minimally resource-intensive. Cost here means total resource cost per investigation, including software, expertise, and computational power, not just hardware cost. While SHM systems have big capital costs but are often deployed on existing infrastructure; in contrast, each computational simulation represents a new, high-intensity analytical effort.

For complexity, models were classified as high if they involved intricate algorithms or extensive calibration; medium if they integrated multiple methods but remained accessible to trained practitioners; and low if they were rule-based, descriptive, or designed for rapid assessment. This approach ensures that the framework reflects real-world constraints and user capabilities, rather than theoretical assumptions.

The strategic framework in Figure 10 represents a novel contribution beyond simple classification. By integrating three key dimensions—complexity, data needs, and cost—it functions as a decision-support tool that guides practitioners in selecting the most fit-for-purpose model. This contrasts with static taxonomies found in prior literature and addresses a critical gap in translating theoretical models into practical application.

6. Conclusions

This review has traced the evolution of risk analysis in structural collapse investigations over the past decade and a half, revealing a field in transition. The dominant trend is no longer the use of isolated models, but the strategic integration of diverse analytical approaches a shift from asking what failed to understanding why it was allowed to fail.

We see a clear movement from descriptive, post-event analysis toward predictive, systemic, and proactive frameworks. Early studies relied heavily on computational models like Finite Element Analysis (FEA) or qualitative case studies, each offering deep but narrow insights. However, since 2016, a new pattern has emerged: the hybridization of methods. Combining FEA with CFD for fire-structure interaction, FTA with Bayesian Networks for probabilistic reasoning, or STAMP with fuzzy DEMATEL for causal mapping reflects a

growing effort to capture the full complexity of structural failures not just the technical, but the human, organizational, and environmental dimensions.

This integration has significantly enhanced the accuracy and comprehensiveness of investigations. For example, the analysis of the Plasco Building collapse combined CFD simulations, FEA, forensic engineering, and systemic models to reconstruct the event from ignition to collapse. Similarly, the investigation of the Morandi Bridge collapse integrated remote sensing, structural monitoring, and material testing to identify both immediate and long-term causes. These multi-method approaches allow for cross-validation, reducing uncertainty and strengthening conclusions.

Yet, this sophistication comes at a cost. The most advanced models are often data-intensive, computationally expensive, and inaccessible in low-resource settings. They require high-quality data, expert personnel, and significant investment, which raises concerns about equity in structural safety. On the other hand, simpler models like FTA or AHP, while more accessible, may oversimplify complex realities and fail to capture systemic interactions. Among the emerging approaches, one stands out for its simplicity, clarity, and historical grounding: the Synergy and Accumulation Model for Analysis (SAMA). Developed to address the limitations of traditional models, SAMA focuses on how small, overlooked risks accumulate and interact over time to produce catastrophic failures. Unlike models that rely on real-time data or complex simulations, SAMA leverages past failures as a foundation for future prevention. It transforms historical data into actionable knowledge, making it both accessible and deeply insightful.

This tension between complexity and practicality is at the heart of modern risk analysis. The choice of model is not a question of superiority, but of fitness for purpose. A model that works for a high-profile bridge collapse may be impractical for a routine building inspection in a developing region. This is where the strategic framework in Figure 10 becomes invaluable not as a ranking of models, but as a decision-support tool that balances complexity, data needs, and cost.

Some of the models have already shown real world application. For example:

- The HYRISK model is being used by the U.S. Federal Highway Administration for infrastructure risk assessment, including bridge safety evaluations.
- Structural Health Monitoring (SHM) systems are being deployed in critical bridges and tunnels worldwide, for continuous assessment and early warning.
- Remote sensing (MT-InSAR) is being used by municipal authorities and engineering firms to monitor ground deformation in urban areas, as seen in the Mexico City Metro and the Morandi Bridge before it collapsed.

Based on this synthesis, we have:

- i. Hybrid Lightweight Models: Future work should focus on creating integrated frameworks that combine the strengths of different quadrants (e.g., FEA + FTA, SHM + AHP) without excessive computational overhead.
- ii. Standardization of Risk Analysis Protocols: We need industry wide standards for model selection, data reporting and validation procedures.
- iii. AI and Digital Twins: How can we use machine learning and digital twin technologies for real-time monitoring, predictive analytics and automated risk assessment.
- iv. Equitable Access to Advanced Tools: How can we make advanced models accessible in low resource contexts through open-source platforms, simplified interfaces and capacity building programs.

Looking ahead, the future of structural safety lies not in the dominance of any single model, but in the intelligent combination of complementary methods. We need hybrid models that balance accuracy and accessibility. We need standardized protocols for data

collection and model validation. And we need to bridge the gap between academia and practice, ensuring that advanced tools are not confined to elite institutions but are available to all who manage our built environment.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/buildings15203701/s1>, Table S1: PRISMA Checklist, demonstrating full adherence to reporting guidelines for systematic reviews. Figure S1: PRISMA Flow Diagram, illustrating the study selection process from initial database search to final inclusion of 101 studies.

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Abbreviations

The following abbreviations are used in this manuscript:

AEM	Applied Element Method
AHP	Analytic Hierarchy Process
AMMP	Accident Mechanism of Multiple Processes
BBN	Bayesian Belief Network
CFD	Computational Fluid Dynamics
CFCFSF	Contributing Factors Classification System Framework
DSA	Driller's Situation Awareness
ETM	Energy Transfer Model
FDS	Fire Dynamics Simulator
FEA	Finite Element Analysis
FDM	Finite Difference Method
FE	Finite Element
FMEA	Failure Mode and Effect Analysis
FTA	Fault Tree Analysis
GIS	Geographic Information System
HFACS	Human Factors Analysis and Classification System
HOFs	Human and Organizational Factors
IDF	Future Degradation Index
IRD	Degree Relevance Index
ISM	Interpretive Structural Modeling
LLM	Large Language Model
LSM	Life Safety Model
MT-InSAR	Multi-Temporal Interferometric Synthetic Aperture Radar
PRA	Probabilistic Risk Assessment
QBDC	Query-by-Dropout-Committee
RBS	Risk Breakdown Structure
RPN	Risk Priority Number
SCM	Swiss Cheese Model
SD	System Dynamics
SAMA	Synergy and Accumulation Model for Analysis
SHM	Structural Health Monitoring
STAMP	Systems-Theoretic Accident Modeling and Processes
WBS	Work Breakdown Structure

References

- Newman, P. Beneath the horizon. *Eng. Technol.* **2010**, *5*, 36–39. [\[CrossRef\]](#)
- Roberts, R.; Flin, R.; Cleland, J. “Everything was fine”*: An analysis of the drill crew’s situation awareness on Deepwater Horizon. *J. Loss Prev. Process Ind.* **2015**, *38*, 87–100. [\[CrossRef\]](#)
- Mohammadfam, I.; Gholamizadeh, K. Investigation of causes of Plasco building accident in Iran using timed MTO and ACCIMAP methods: Investigation of Plasco 4 building accident in Iran. *J. Fail. Anal. Prev.* **2020**, *20*, 2087–2096. [\[CrossRef\]](#)
- Elwood, K.J. Performance of concrete buildings in the 22 February 2011 Christchurch earthquake and implications for Canadian codes. *Can. J. Civ. Eng.* **2013**, *40*, 759–776. [\[CrossRef\]](#)
- Kam, W.Y.; Pampanin, S.; Elwood, K. Seismic performance of reinforced concrete buildings in the 22 February Christchurch (Lyttleton) earthquake. *Bull. N. Z. Soc. Earthq. Eng.* **2011**, *44*, 239–278.
- Ataei, H.; Niazy, A.-S.M. Buckling analysis of arched structures using finite element analysis. In *Forensic Engineering 2015*; ASCE Library: Reston, VA, USA, 2015; pp. 824–834.
- Manzur, T.; Mahmood, M.H.; Baten, B.; Hasan, M.J.; Hossain, M.R.; Noor, M.A.; Yazdani, N. Assessment of progressive collapse proneness of existing typical garment factory buildings in Bangladesh. *J. Perform. Constr. Facil.* **2020**, *34*, 04020092. [\[CrossRef\]](#)
- Stark, T.D.; Benekohal, R.; Fahnstock, L.A.; LaFave, J.M.; He, J.; Wittenkeller, C. I-5 Skagit river bridge collapse review. *J. Perform. Constr. Facil.* **2016**, *30*, 04016061. [\[CrossRef\]](#)
- Liu, R.; Nastar, N.; Cavalline, T.; Sullivan-Green, L.E.; Bosela, P.A.; Delatte, N.J.; Parfitt, M.K.; Carper, K.L. Failing Forward—Construction Failure Case Studies. In *Eighth Congress on Forensic Engineering*; American Society of Civil Engineers: Reston, VA, USA, 2018.
- Cao, R.; Agrawal, A.K.; El-Tawil, S. Overheight impact on bridges: A computational case study of the Skagit River bridge collapse. *Eng. Struct.* **2021**, *237*, 112215. [\[CrossRef\]](#)
- Cook, W.; Barr, P.J.; Halling, M.W. Bridge failure rate. *J. Perform. Constr. Facil.* **2015**, *29*, 04014080. [\[CrossRef\]](#)
- Ghasemi, S.H.; Nowak, A.S.; Nazari, R. Analyzing reliability missteps: The collapse of the Francis Scott Key Bridge—Target reliability, redundancy, and extreme load combinations. In *Structures*; Elsevier: Amsterdam, The Netherlands, 2025.
- Ewa, D.E. Building collapse in south-south Nigeria. In Proceedings of the Standards Organization of Nigeria (SON) Workshop on Increased Awareness of Building Collapse in Nigeria, South-South Zone, Calabar, Nigeria, 15 March 2018.
- Caicedo, B.; Alonso, E.; Mendoza, C.; Alcoverro, J. The collapse of Space building. *Géotechnique* **2019**, *69*, 260–273. [\[CrossRef\]](#)
- Awe, O.O.; Atofarati, E.O.; Adeyinka, M.O.; Musa, A.P.; Onasanya, E.O. Assessing the factors affecting building construction collapse casualty using machine learning techniques: A case of Lagos, Nigeria. *Int. J. Constr. Manag.* **2024**, *24*, 261–269. [\[CrossRef\]](#)
- Zhang, H.; Yu, D.; Li, G.; Dong, Z. A real-time seismic damage prediction framework based on machine learning for earthquake early warning. *Earthq. Eng. Struct. Dyn.* **2024**, *53*, 593–621. [\[CrossRef\]](#)
- Lazaridis, P.C.; Kavvadias, I.E.; Demertzis, K.; Iliadis, L.; Vasiliadis, L.K. Structural damage prediction of a reinforced concrete frame under single and multiple seismic events using machine learning algorithms. *Appl. Sci.* **2022**, *12*, 3845. [\[CrossRef\]](#)
- Wen, W.; Hu, J.; Zhai, C.; Pei, S. Post-earthquake functionality assessment of subway stations considering the interdependency among sub-systems. *Comput.-Aided Civ. Infrastruct. Eng.* **2024**, *39*, 136–161. [\[CrossRef\]](#)
- Yan, Y.; Huang, Q.; Xie, Y.; Qiu, J.; Han, Y.; Teng, H.; Liu, T. Failure analysis and deformation mechanism of segmented utility tunnels crossing ground fissure zones with different intersection angles. *Eng. Fail. Anal.* **2022**, *139*, 106456. [\[CrossRef\]](#)
- Leu, S.-S.; Chang, C.-M. Bayesian-network-based safety risk assessment for steel construction projects. *Accid. Anal. Prev.* **2013**, *54*, 122–133. [\[CrossRef\]](#)
- Rotta, L.H.S.; Alcantara, E.; Park, E.; Negri, R.G.; Lin, Y.N.; Bernardo, N.; Mendes, T.S.G.; Souza Filho, C.R. The 2019 Brumadinho tailings dam collapse: Possible cause and impacts of the worst human and environmental disaster in Brazil. *Int. J. Appl. Earth Obs. Geoinf.* **2020**, *90*, 102119.
- Zhang, Z.; Chu, X.; Abbasi, R.; Ghisbain, P.; Cao, L.; Abruzzo, J. Analysis of Hurricane Maria’s Impact on the Arecibo Telescope. *J. Perform. Constr. Facil.* **2025**, *39*, 04025021. [\[CrossRef\]](#)
- Hartono, W.; Kristiawan, S.A.; Handayani, D.; Sutopo, W. Safety Risk Assessment Model for Bridge Construction. *Civ. Eng. J.* **2025**, *11*, 161–184. [\[CrossRef\]](#)
- Domada, R.; Khan, A.A.; Orabi, A.; Usmani, A. Analysis of the plasco tower in fire using an integrated simulation approach. *Fire Saf. J.* **2025**, 104459. [\[CrossRef\]](#)
- Ren, X.; Du, Z.; Wang, J.; Yang, F.; Su, T.; Wei, W. Safety decision analysis of collapse accident based on “accident tree–analytic hierarchy process”. *Nonlinear Eng.* **2023**, *12*, 20220295. [\[CrossRef\]](#)
- Nacheman, S.G.; Jackson, K.J.; Nelson, J.K.; Pinto, M.J.; Valderruten, L.; Bhagath, A. The Indiana state fair collapse incident: Anatomy of a failure. In *Forensic Engineering 2012: Gateway to a Safer Tomorrow*; American Society Civil Engineers: Reston, VA, USA, 2013; pp. 1070–1080.
- Ale, B.; Van Gulijk, C.; Hanea, A.; Hanea, D.; Hudson, P.; Lin, P.-H.; Sillem, S. Towards BBN based risk modelling of process plants. *Saf. Sci.* **2014**, *69*, 48–56. [\[CrossRef\]](#)

28. Borsje, H.; Renier, B.; Burggraaf, H. Collapse of the roof of a football stadium. In Proceedings of the IABSE Symposium: Engineering for Progress, Nature and People, Madrid, Spain, 3–5 September 2014.
29. Chao, F.; Yu, L.; Huo, J.X.; Wang, M.N. Risk analysis of collapse during construction for a subway transfer station with large span and small clearance. *Appl. Mech. Mater.* **2014**, *584*, 2077–2082. [\[CrossRef\]](#)
30. Pranesh, V.; Palanichamy, K.; Saidat, O.; Peter, N. Lack of dynamic leadership skills and human failure contribution analysis to manage risk in deep water horizon oil platform. *Saf. Sci.* **2017**, *92*, 85–93. [\[CrossRef\]](#)
31. Weng, J.; Tan, K.H.; Lee, C.K. Modeling progressive collapse of 2D reinforced concrete frames subject to column removal scenario. *Eng. Struct.* **2017**, *141*, 126–143. [\[CrossRef\]](#)
32. Aires, U.R.V.; Santos, B.S.M.; Coelho, C.D.; da Silva, D.D.; Calijuri, M.L. Changes in land use and land cover as a result of the failure of a mining tailings dam in Mariana, MG, Brazil. *Land Use Policy* **2018**, *70*, 63–70. [\[CrossRef\]](#)
33. Akpabot, A.I.; Ede, A.N.; Olofinnade, O.M.; Bamigboye, G. Predicting buildings collapse due to seismic action in Lagos state. *Int. J. Eng. Res. Afr.* **2018**, *37*, 91–102. [\[CrossRef\]](#)
34. Akp, A. Forecasting the hazards of seismic induced building collapse in Lagos Nigeria through quality of reinforcing steel bars. *Int. J. Mech. Eng. Technol.* **2018**, *9*, 766–775.
35. Yarlagaadda, T.; Hajiloo, H.; Jiang, L.; Green, M.; Usmani, A. Preliminary modelling of Plasco Tower collapse. *Int. J. High-Rise Build.* **2018**, *7*, 397–408.
36. Shakib, H.; Pirizadeh, M.; Dardaei, S.; Zakersalehi, M. Technical and administrative assessment of Plasco building incident. *Int. J. Civ. Eng.* **2018**, *16*, 1227–1239. [\[CrossRef\]](#)
37. Bazzucchi, F.; Restuccia, L.; Ferro, G.A. Considerations over the Italian road bridge infrastructure safety after the Polcevera viaduct collapse: Past errors and future perspectives. *Fract. Struct. Integr./Frat. Ed Integrità Strutt.* **2018**, *12*, 400–421. [\[CrossRef\]](#)
38. Wu, G.; Qiu, W.; Wu, T. Nonlinear dynamic analysis of the self-anchored suspension bridge subjected to sudden breakage of a hanger. *Eng. Fail. Anal.* **2019**, *97*, 701–717. [\[CrossRef\]](#)
39. Di Prisco, M. Critical infrastructures in Italy: State of the art, case studies, rational approaches to select the intervention priorities. In Proceedings of the Fib Symposium 2019: Concrete-Innovations in Materials, Design and Structures, Krakow, Poland, 27–29 May 2019; International Federation for Structural Concrete: Lausanne, Switzerland, 2019.
40. Zhou, X.; Di, J.; Tu, X. Investigation of collapse of Florida International University (FIU) pedestrian bridge. *Eng. Struct.* **2019**, *200*, 109733. [\[CrossRef\]](#)
41. Jiang, L.; Ye, J.; Zheng, H. Collapse mechanism analysis of the FIU pedestrian bridge based on the improved structural vulnerability theory (ISVT). *Eng. Fail. Anal.* **2019**, *104*, 1064–1075. [\[CrossRef\]](#)
42. Milillo, P.; Giardina, G.; Perissin, D.; Milillo, G.; Coletta, A.; Terranova, C. Pre-collapse space geodetic observations of critical infrastructure: The Morandi Bridge, Genoa, Italy. *Remote Sens.* **2019**, *11*, 1403. [\[CrossRef\]](#)
43. Cambridge, M.; Shaw, D. Preliminary reflections on the failure of the Brumadinho tailings dam in January 2019. *Dams Reserv.* **2019**, *29*, 113–123. [\[CrossRef\]](#)
44. Malomo, D.; Scattarreggia, N.; Orgnoni, A.; Pinho, R.; Moratti, M.; Calvi, G.M. Numerical study on the collapse of the Morandi bridge. *J. Perform. Constr. Facil.* **2020**, *34*, 04020044. [\[CrossRef\]](#)
45. Wu, G.; Qiu, W. Effect of load cases and hanger-loss scenarios on dynamic responses of the self-anchored suspension bridge to abrupt rupture of hangers. *J. Perform. Constr. Facil.* **2020**, *34*, 04020081. [\[CrossRef\]](#)
46. Boateng, F.G. Exploring the collapse of buildings in urban settings. *Proc. Inst. Civ. Eng.-Munic. Eng.* **2020**, *173*, 187–195. [\[CrossRef\]](#)
47. Putcha, C.; Dutta, S.; Rodriguez, J. Risk priority number for bridge failures. *Pract. Period. Struct. Des. Constr.* **2020**, *25*, 04020010. [\[CrossRef\]](#)
48. Shakib, H.; Zakersalehi, M.; Jahangiri, V.; Zamanian, R. Evaluation of Plasco Building fire-induced progressive collapse. In *Structures*; Elsevier: Amsterdam, The Netherlands, 2020.
49. Ahmadi, M.; Aghakouchak, A.; Mirghaderi, R.; Tahouni, S.; Garivani, S.; Shahmari, A.; Epackachi, S. Collapse of the 16-story Plasco building in Tehran due to fire. *Fire Technol.* **2020**, *56*, 769–799. [\[CrossRef\]](#)
50. Clemente, P. Monitoring and evaluation of bridges: Lessons from the Polcevera Viaduct collapse in Italy. *J. Civ. Struct. Health Monit.* **2020**, *10*, 177–182. [\[CrossRef\]](#)
51. Morgese, M.; Ansari, F.; Domaneschi, M.; Cimellaro, G.P. Post-collapse analysis of Morandi's Polcevera viaduct in Genoa Italy. *J. Civ. Struct. Health Monit.* **2020**, *10*, 69–85. [\[CrossRef\]](#)
52. Domaneschi, M.; Pellicchia, C.; De Iuliis, E.; Cimellaro, G.; Morgese, M.; Khalil, A.; Ansari, F. Collapse analysis of the Polcevera viaduct by the applied element method. *Eng. Struct.* **2020**, *214*, 110659. [\[CrossRef\]](#)
53. Imafidon, M.; Ogbu, C. A taxonomy of building collapse causes in Lagos State Nigeria. *Niger. J. Technol.* **2020**, *39*, 74–86. [\[CrossRef\]](#)
54. Sediek, O.A.; El-Tawil, S.; McCormick, J. Seismic debris field for collapsed RC moment resisting frame buildings. *J. Struct. Eng.* **2021**, *147*, 04021045. [\[CrossRef\]](#)
55. Shen, G.; Lu, Y.; Zhang, S.; Xiang, Y.; Sheng, J.; Fu, J.; Fu, S.; Liu, M. Risk dynamics modeling of reservoir dam break for safety control in the emergency response process. *Water Supply* **2021**, *21*, 1356–1371. [\[CrossRef\]](#)

56. Primo, P.P.B.; Antunes, M.N.; Ramos, M.P.; Arias, A.R.L.; Oliveira, A.E.; Siqueira, C.E. Mining disasters in Brazil: A case study of dam ruptures in Mariana and Brumadinho. *Case Stud. Environ.* **2021**, *5*, 1242438. [[CrossRef](#)]
57. Adeyemi, H.O.; Alaba, S.T.; Odugboise, B.D.; Babalola, A.A. Impact assessment of mechanical services to building instability. *Proc. Eng.* **2021**, *3*, 433–440.
58. Hu, Y.-C.; Tan, Y.-H.; Xi, F. Failure assessment and virtual scenario reproduction of the progressive collapse of the FIU bridge. *Eng. Struct.* **2021**, *227*, 111423. [[CrossRef](#)]
59. Rymasz, J. Causes of the collapse of the Polcevera Viaduct in Genoa, Italy. *Appl. Sci.* **2021**, *11*, 8098. [[CrossRef](#)]
60. Cheng, D.; Cui, Y.; Li, Z.; Iqbal, J. Watch out for the tailings pond, a sharp edge hanging over our heads: Lessons learned and perceptions from the brumadinho tailings dam failure disaster. *Remote Sens.* **2021**, *13*, 1775. [[CrossRef](#)]
61. Huang, Y.-J.; Tao, J.; Yang, F.-Q.; Chen, C. Construction safety during pandemics: Learning from the Xinjia Express hotel collapse during COVID-19 in China. *Int. J. Environ. Res. Public Health* **2021**, *18*, 11498. [[CrossRef](#)] [[PubMed](#)]
62. Goncalves Filho, A.P.; Waterson, P.; Jun, G.T. Improving accident analysis in construction—Development of a contributing factor classification framework and evaluation of its validity and reliability. *Saf. Sci.* **2021**, *140*, 105303. [[CrossRef](#)]
63. Wang, H.; Chen, Q.; Agrawal, A.K.; El-Tawil, S.; Bhattacharya, B.; Wong, W. Dynamic response and progressive collapse of a long-span suspension bridge induced by suspender loss. *J. Struct. Eng.* **2022**, *148*, 05022001. [[CrossRef](#)]
64. Oswald, D.; Moore, T. *Constructing a Consumer-Focused Industry: Cracks, Cladding and Crisis in the Residential Construction Sector*; Routledge: London, UK, 2022.
65. Khan, A.A.; Domada, R.V.V.; Huang, X.; Khan, M.A.; Usmani, A. Modeling the collapse of the Plasco Building. Part I: Reconstruction of fire. *Build. Simul.* **2022**, *15*, 583–596. [[CrossRef](#)]
66. Epackachi, S.; Mirghaderi, S.R.; Aghelizadeh, P. Failure analysis of the 16-story Plasco building under fire condition. *Sci. Iran.* **2022**, *29*, 1107–1124. [[CrossRef](#)]
67. Won, J.-H.; Jeong, H.-J.; Kim, W.; Kim, S.; Kang, S.-Y.; Hwang, J.M. Mechanisms analysis for fatal accident types caused by multiple processes in the workplace: Based on accident case in South Korea. *Int. J. Environ. Res. Public Health* **2022**, *19*, 11430. [[CrossRef](#)]
68. Scattarreggia, N.; Salomone, R.; Moratti, M.; Malomo, D.; Pinho, R.; Calvi, G.M. Collapse analysis of the multi-span reinforced concrete arch bridge of Caprigliola, Italy. *Eng. Struct.* **2022**, *251*, 113375. [[CrossRef](#)]
69. Zhang, Y.; Xing, X.; Antwi-Afari, M.F.; Wu, M. Safety risk estimation of construction project based on energy transfer model and system dynamics: A case study of collapse accident in China. *Int. J. Environ. Res. Public Health* **2022**, *19*, 14386. [[CrossRef](#)]
70. Zhou, Z.; Goh, Y.M.; Shi, Q.; Qi, H.; Liu, S. Data-driven determination of collapse accident patterns for the mitigation of safety risks at metro construction sites. *Tunn. Undergr. Space Technol.* **2022**, *127*, 104616. [[CrossRef](#)]
71. Okur, E.K.; Okur, F.Y.; Altunışık, A.C.; Günaydin, M.; Adanur, S. Progressive collapse assessment of Osmangazi Suspension Bridge due to sudden hanger breakage under different loading conditions. *Eng. Fail. Anal.* **2023**, *149*, 107269. [[CrossRef](#)]
72. Caredda, G.; Makoon, N.; Buitrago, M.; Sagaseta, J.; Chrysanthopoulos, M.; Adam, J.M. Learning from the progressive collapse of buildings. *Dev. Built Environ.* **2023**, *15*, 100194. [[CrossRef](#)]
73. Choudhry, R.M.; Zafar, I.; Almatawa, M.S. Proactive approach to measure safety management on building projects in Saudi Arabia. *Adv. Civ. Eng.* **2023**, *2023*, 6306157. [[CrossRef](#)]
74. Farouk, M. Structural and environmental safety studies of the holy mosque area using CFD. *Buildings* **2023**, *13*, 1809. [[CrossRef](#)]
75. Rose, R.L.; Mugi, S.R.; Saleh, J.H. Accident investigation and lessons not learned: AcciMap analysis of successive tailings dam collapses in Brazil. *Reliab. Eng. Syst. Saf.* **2023**, *236*, 109308. [[CrossRef](#)]
76. Okoye, J.; Apeh, S.; Olaye, E.; Osuji, S. Towards automation of building integrity tracking: Review of building collapse in Nigeria. *NIPES-J. Sci. Technol. Res.* **2023**, *5*, 179–194.
77. Khan, A.A.; Khan, M.A.; Domada, R.V.V.; Huang, X.; Usmani, A.; Bakhtiyari, S.; Ashtiani, M.J.; Garivani, S.; Aghakouchak, A.A. Fire modelling framework for investigating tall building fire: A case study of the Plasco Building. *Case Stud. Therm. Eng.* **2023**, *45*, 103018. [[CrossRef](#)]
78. Lumbroso, D.; Davison, M.; Wetton, M. Development of an agent-based model to improve emergency planning for floods and dam failures. *J. Hydroinformatics* **2023**, *25*, 1610–1628. [[CrossRef](#)]
79. Chen, X.; Qiao, W. A hybrid STAMP-fuzzy DEMATEL-ISM approach for analyzing the factors influencing building collapse accidents in China. *Sci. Rep.* **2023**, *13*, 19745. [[CrossRef](#)] [[PubMed](#)]
80. Rodríguez-Antuñano, I.; Martínez-Sánchez, J.; Cabaleiro, M.; Riveiro, B. Anticipating the collapse of urban infrastructure: A methodology based on earth observation and MT-InSAR. *Remote Sens.* **2023**, *15*, 3867. [[CrossRef](#)]
81. Logan, L.; Singh, A. High-rise condominium collapse. *J. Leg. Aff. Disput. Resolut. Eng. Constr.* **2023**, *15*, 05022008. [[CrossRef](#)]
82. Ma, D.; Zhang, C.; Zhao, L.; Huang, Q.; Liu, B. An analysis of the evolution of public sentiment and spatio-temporal dynamics regarding building collapse accidents based on Sina Weibo data. *ISPRS Int. J. Geo-Inf.* **2023**, *12*, 388. [[CrossRef](#)]
83. Whitehead, B. Tretten Bridge: A Numerical Study into the Collapse of a Steel-Glulam Structure. 2023. Available online: https://oro.open.ac.uk/96014/2/WHITEHEAD_T460_RVOR.pdf (accessed on 6 June 2025).

84. Anastasiadis, A.; Sachpazis, C. Failure analysis of the FC Twente Stadium roof collapse using the Swiss Cheese Accident Model. In Proceedings of the 10th Hellenic Conference on Metal Structures, Paper Number of Reference, Athens, Greece, 19–21 October 2023.
85. Chen, Q.; Wang, H.; Bhattacharya, B.; El-Tawil, S.; Agrawal, A.K.; Wong, W. Reliability-based framework for structural robustness evaluation of bridges. *J. Bridge Eng.* **2024**, *29*, 04024035. [\[CrossRef\]](#)
86. Zheng, X.; Jian, J.; Liu, L.; Sun, B.; Zhang, K.; Gao, H. Static and dynamic performance analysis of cable-stayed bridges with cables damaged fire. *Buildings* **2024**, *14*, 884. [\[CrossRef\]](#)
87. Suarjana, M.; Mani'Sarungallo, A. Investigation of the Collapse of The Cincin Lama Bridge with Consideration of Fatigue Damage. *GEOMATE J.* **2024**, *26*, 100–107. [\[CrossRef\]](#)
88. Wang, J.; Zhang, L.; Yang, H.; Liu, H.; Skibniewski, M.J. Dynamic reliability analysis of Aerial Building Machine under extreme wind loads using improved QBDC-based active learning. *Reliab. Eng. Syst. Saf.* **2024**, *244*, 109927. [\[CrossRef\]](#)
89. Tang, D.; Huang, M. The sustainable development of bridges in China: Collapse cause analysis, existing management dilemmas and potential solutions. *Buildings* **2024**, *14*, 419. [\[CrossRef\]](#)
90. Hao, Y.; Qi, J.; Ma, X.; Wu, S.; Liu, R.; Zhang, X. An LLM-based inventory construction framework of urban ground collapse events with spatiotemporal locations. *ISPRS Int. J. Geo-Inf.* **2024**, *13*, 133. [\[CrossRef\]](#)
91. Guibaud, A.; Mindeguia, J.-C.; Albuerne, A.; Parent, T.; Torero, J. Notre-Dame de Paris as a validation case to improve fire safety modelling in historic buildings. *J. Cult. Herit.* **2024**, *65*, 145–154. [\[CrossRef\]](#)
92. Solano-Rojas, D.; Wdowinski, S.; Cabral-Cano, E.; Osmanoglu, B. Geohazard assessment of Mexico City's Metro system derived from SAR interferometry observations. *Sci. Rep.* **2024**, *14*, 6035. [\[CrossRef\]](#)
93. Tang, D.; Huang, M. Dilemmas and Solutions for Sustainability-Based Engineering Ethics: Lessons Learned from the Collapse of a Self-Built House in Changsha, Hunan, China. *Buildings* **2024**, *14*, 2581. [\[CrossRef\]](#)
94. Saldaña Cabanillas, A.; Lupaca Huarac, M. Threats to the preservation of the viceregal church facades of Juli, Peru. *J. Inst. Conserv.* **2024**, *47*, 49–67. [\[CrossRef\]](#)
95. Shukla, R.; Pabbisetty, S.K.; Jayanthi, S.; Janardhanan, K. Evaluating the damage of collapsed bridges using remote sensing technologies: Case study: Baltimore's Francis Scott Key Bridge. *Build. Eng.* **2024**, *2*, 1811. [\[CrossRef\]](#)
96. Qi, H.; Zhou, Z.; Irizarry, J.; Deng, X.; Yang, Y.; Li, N.; Zhou, J. Modification of HFACS model for path identification of causal factors of collapse accidents in the construction industry. *Eng. Constr. Archit. Manag.* **2024**, *32*, 4718–4745. [\[CrossRef\]](#)
97. Medaa, E.; Shirzadi Javid, A.A.; Malekitabar, H.; Banihashemi, S. The Synergy and Accumulation Model for Analysis (SAMA): A Novel Approach to Transforming Risk Analysis in Construction with a Focus on the Deepwater Horizon Disaster Case. *Buildings* **2025**, *15*, 1879. [\[CrossRef\]](#)
98. Murphy, T.P.; Galindez, N.Y.; Artmont, F.A.; Adams, A.R.; Lopez de Murphy, M. Collapse of the Chirajara Cable-Stayed Bridge in Colombia. *Transp. Res. Rec.* **2025**, *2679*, 104–121. [\[CrossRef\]](#)
99. D'Angelo, M.; Civera, M.; Giordano, P.F.; Borlenghi, P.; Ballio, F.; Limongelli, M.P.; Chiaia, B. Bridge collapses in Italy across the 21st century: Survey and statistical analysis. *Struct. Infrastruct. Eng.* **2025**, 1–23. [\[CrossRef\]](#)
100. Cros, S.; Tiberghien, B.; Bertolucci, M. Senselisting and the Reorganisation of Collective Action During Crisis Management: The Notre-Dame de Paris fire. *Organ. Stud.* **2025**, *46*, 941–965. [\[CrossRef\]](#)
101. Porcu, M.C.; Buitrago, M.; Calderón, P.A.; Garau, M.; Cocco, M.F.; Adam, J.M. Robustness-based assessment and monitoring of steel truss railway bridges to prevent progressive collapse. *J. Constr. Steel Res.* **2025**, *226*, 109200. [\[CrossRef\]](#)
102. Klym, A.; Blikharsky, Y.; Gunka, V.; Poliak, O.; Selejdak, J.; Blikharsky, Z. An Overview of the Main Types of Damage and the Retrofitting of Reinforced Concrete Bridges. *Sustainability* **2025**, *17*, 2506. [\[CrossRef\]](#)
103. Shahzada, K.; Noor, E.U.A.; Xu, Z.-D. China-Pakistan Belt and Road Joint Laboratory on Smart Disaster Prevention of Major Infrastructures. 2025. Available online: https://www.researchgate.net/profile/Umar-Ahmad-Noor-2/publication/390549998_In_the_Wake_of_the_March_28_2025_Myanmar_Earthquake_A_Detailed_Examination/links/67f385879b1c6c4877805207/In-the-Wake-of-the-March-28-2025-Myanmar-Earthquake-A-Detailed-Examination.pdf (accessed on 3 October 2025).

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